



Physics-constrained neural networks for high-fidelity composite failure envelopes

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ABSTRACT

To reconcile classical theory with data-driven modeling, this study proposes a Physics-Constrained Neural Network (PCNN) for the prediction of composite multi-axial failure. Validated across 16 distinct test cases covering carbon and glass fiber systems under complex biaxial and triaxial stress states, the framework integrates established failure criteria—including Tsai-Wu, Cuntze, and LaRC—through an adaptive weighting mechanism that helps balance data fit and physical consistency. A novel decoupled prediction strategy is employed, transforming the high-dimensional stress tensor prediction into a scalar regression problem while preserving essential directional information. Furthermore, the adaptive weighting mechanism is introduced to dynamically balance the gradients between data fidelity and physical consistency during training. Results show that the PCNN outperforms the purely data-driven Artificial Neural Network (ANN) baseline. Specifically, the model incorporating high-fidelity LaRC constraints achieves a Coefficient of Determination of 0.9198, compared to the unconstrained ANN ($R^2 = 0.8660$), suggesting a positive association between the fidelity of the inductive bias and predictive accuracy. In addition, the PCNN framework effectively suppresses the heavy-tailed error distribution typical of black-box models and reconstructs physically consistent, closed failure envelopes that rectify local biases in traditional theories. This work offers a reliable, physically consistent, and practical tool with improved accuracy for composite design, demonstrating the synergy between domain knowledge and machine learning.

1. Introduction

A foundational task in composite structure design is defining the failure envelope—the boundary in stress space that separates safe from critical states [1]. However, the theoretical elegance of this concept belies a persistent practical challenge. Real-world structures, from torsion-loaded fuselages to bending wings [2,3], operate under stress states that interact in complex, often non-intuitive ways, rendering simplistic failure criteria unreliable [4]. This gap between a deterministic design goal and the unpredictable nature of multiaxial failure underscores the need for more sophisticated predictive models [5,6].

To this end, the need for accurate failure models has historically driven the pursuit of two distinct paths [2,7]: micromechanical modeling, which seeks to resolve failure initiation at the constituent level, and macroscopic strength theories, which phenomenologically describe the failure envelope at the laminate scale. Consequently, macroscopic strength theories have emerged as the pragmatic choice for engineering design due to their direct applicability. These theories are

primarily divided between those that rely solely on phenomenological descriptions and those that explicitly account for underlying physical mechanisms. The former, like the Tsai-Wu criterion [8], treats the composite as a homogeneous continuum and predicts failure using a unified function without distinguishing specific failure modes. In contrast, the latter establishes independent failure criteria for different constituents (e.g., fibers and matrix), thereby attempting to capture the underlying physical mechanisms of failure. Such physically-grounded models, exemplified by the Cuntze [9] and LaRC criteria [10–12], derive their formulation from the principles of continuum mechanics [1]. A key advantage of this approach is that their parameters are directly linked to experimentally measurable material properties such as uniaxial strengths, which not only reduces phenomenological fitting but also provides them with clear physical interpretability. Despite their interpretability advantages, applying these classical criteria to complex multi-axial stress states remains challenging [13,14]. A primary limitation stems from their insufficient consideration of the coupling effects among different stress components.

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This insufficiency in modeling coupling effects becomes evident when examining specific loading scenarios. For example, under longitudinal–transverse biaxial loading, the original Hashin criterion [15] predicts a rectangular failure envelope [16]. This prediction shows a discrepancy with the elliptical envelopes observed experimentally, particularly in the tension–compression quadrant [17]. This discrepancy highlights two key aspects of such criteria. Functionally, they define a failure surface to assess the safety of a given stress state, rather than directly predicting the actual failure strength under a specific loading direction. Physically, these models often simplify complex damage mechanisms into a superposition of independent modes for mathematical conciseness. This approach overlooks the physical interactions between stress components, such as the enhancing effect of transverse compression on the matrix shear strength [18]. Consequently, their predictive accuracy is limited when characterizing the true coupled damage mechanisms under multi-axial loading. The challenge of explicitly formulating the failure function has led researchers to explore data-driven alternatives. Here, Artificial Neural Networks (ANNs) present a natural solution [19–22]. These models learn from extensive sets of input–output data to construct nonlinear mappings from material properties and structural parameters to failure states and damage evolution processes [23,24]. Recent advances have demonstrated the efficacy of deep learning in complex tasks [25–27]. This process effectively bypasses the need for complex physical modelling [28,29]. Despite their powerful capabilities in nonlinear fitting [30–32], purely data-driven methods have inherent drawbacks. Their predictive performance is highly dependent on the completeness and balance of the training data [33–35], thereby exhibiting poor generalizability to unseen material systems. Furthermore, their predictions can be unstable and physically inconsistent when applied to complex multi-axial stress combinations where experimental data is sparse [36]. More importantly, the inherent black box nature of neural networks is a major challenge [37]. Their internal weights and biases do not correspond to specific physical quantities. As a result, their decision-making logic is opaque and lacks physical interpretability [38].

In light of these challenges, a growing recognition of the need to integrate domain knowledge with data-driven learning has established the Physics-Guided Machine Learning (PGML) framework as a pivotal research direction [39,40]. Currently, there are several technical paths to achieve this integration [41]. One common approach relies on observation-based constraints, where high-quality data that implicitly contains physical laws is used for training [42]. However, the effectiveness of this approach is dependent on the volume and quality of the data, which limits its applicability. Another path involves architecture-based constraints, also known as inductive biases. This method designs specialized network architectures that inherently satisfy certain physical laws, such as conservation principles or symmetries [43]. While this approach is more fundamental, its implementation is often complex and is typically applicable only to a subset of known principles [39]. In addition, learning-based constraints offer a more versatile pathway for constructing PGML frameworks. The core idea is to modify the optimization objective, particularly the loss function, to incorporate physical knowledge and guide the model toward physically consistent solutions [44]. This flexible approach has become a prominent research direction within the PGML field [41].

The efficacy of this physics-infused paradigm has been demonstrated in recent computational mechanics literature, providing a compelling precedent for constitutive modeling. Notably, researchers have successfully integrated domain knowledge through complementary strategies. On one hand, Roy et al. [45] and Roy and Guha [46] establish physics-augmented loss functions that enforce equilibrium and compatibility conditions. Extending this optimization-centric approach to highly complex material behaviors, the non-associative Drucker–Prager model was recently solved by embedding constitutive laws directly into the loss landscape [47]. On the other hand, addressing the requirement of coordinate independence, Bose and Roy [48] illustrated

the power of architectural inductive biases by embedding Galilean invariance into neural networks for turbulent flows. These advancements collectively confirm that guiding data-driven models with physical principles—whether through loss constraints or specialized architectures—enhances robustness in data-sparse regimes, motivating the adoption of algebraic physical constraints for the anisotropic failure challenges addressed in this work. Furthermore, recent reviews by Karniadakis et al. [41] and Willard et al. [39] emphasize that an advantage of the physics-informed paradigm lies in its capability to handle “imperfect physical models.” In engineering scenarios where analytical theories involve simplifications—such as neglecting in-situ effects in multidirectional laminates—data-driven components can serve as a correction mechanism. By treating physical laws as soft regularizers rather than hard constraints, the neural network effectively engages in a hybrid modeling process, learning the residual discrepancy between the idealized theory and the complex experimental reality. This approach allows the framework to leverage domain knowledge for generalization while retaining the flexibility to calibrate theoretical biases against observational data.

Within the learning-based constraint approach, a widely adopted strategy is to introduce physical constraints directly into the loss function. During training, such models aim not only to minimize the prediction error on observed data, which constitutes the data loss, but also to minimize an additional penalty term [49]. This term, the physical loss, quantifies the extent to which the model's predictions violate known physical laws. These laws can take various forms. One category involves using partial differential equations as constraints [44], leveraging the neural network as a universal function approximator. For example, Wang et al. [50] transformed the classical lamination plate theory into an energy-based loss function to solve for the deformation of composite plates. Alternatively, algebraic or empirical physical relationships offer another mainstream and flexible form of constraint. This approach is powerful for engineering problems where the governing PDEs are complex or intractable, but well-established macroscopic physical principles exist. It is this latter category of constraints—rooted in explicit, algebraic failure criteria—that provides the foundational methodology for the present study, offering a direct pathway to enforce physical consistency without relying on differential operators. For instance, Wang et al. [51] incorporated the empirical Paris' law for crack propagation as a physical constraint in the loss function, thereby enforcing model predictions to adhere to the physical trend of delamination growth. Similarly, in predicting the performance of metal-ceramic composites, Xiong et al. [52] converted principles such as mass conservation, monotonic performance trends based on composition, and the hardness-toughness trade-off into soft constraints within the loss function. While integrating non-differential physical constraints into loss functions has proven effective across various fields [39], its potential remains notably untapped for predicting composite failure under multi-axial stress. This specific problem, characterized by the limitations of classical criteria, therefore presents a promising opportunity to leverage physics-guided learning for achieving a notable improvement in predictive accuracy.

To mitigate the limitations of purely data-driven approaches, this study proposes a Physics-Constrained Neural Network (PCNN) framework designed for high-fidelity failure prediction, as schematically illustrated in Fig. 1. As depicted in Fig. 1a, this framework combines the strengths of traditional physics-based criteria and data-driven models. The proposed framework employs a decoupled prediction strategy (Fig. 1c) to preserve directional information, supplemented by an adaptive weighting mechanism (Fig. 1d) that integrates classical failure criteria as soft constraints. This approach transforms the complex tensor prediction problem into a manageable scalar regression while dynamically enforcing physical consistency during training. Expanded benchmarking against widely established machine learning algorithms demonstrates the efficacy of this approach. Specifically, the proposed PCNN-LaRC model achieves a Coefficient of Determination of 0.9198,

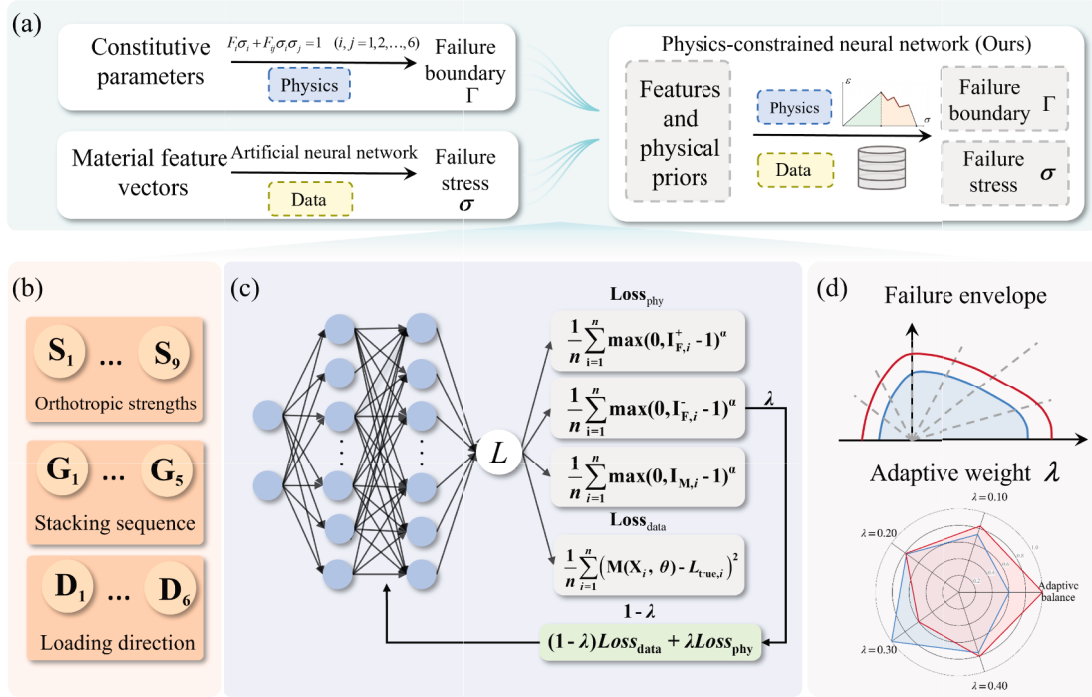


Fig. 1. Framework of the Physics-Constrained Neural Network (PCNN). (a) Schematic comparison of modeling paradigms. The PCNN framework integrates physical priors and data features to simultaneously reconstruct the failure boundary (Γ) and predict failure stress (σ). (b) Input feature space comprising orthotropic strengths (S), stacking sequence (G), and normalized loading direction (D). (c) Network architecture utilizing the decoupled prediction strategy with a hybrid loss function to enforce physical constraints. (d) Adaptive weighting mechanism for dynamically balancing data fidelity and physical consistency to reconstruct closed failure envelopes.

surpassing well-performing baselines such as XGBoost ($R^2 = 0.8769$) and Random Forest ($R^2 = 0.8761$) on the same dataset. Beyond numerical improvements, the framework effectively suppresses the heavy-tailed error distribution typical of black-box models and reconstructs physically consistent failure envelopes, thereby offering an interpretable and precise tool for composite structure design.

2. Multi-axial failure database construction

2.1. Experimental test cases data

To construct a benchmark dataset, we integrate three publicly available research works. These include the first World-Wide Failure Exercise (WWFE-I) [53–55], the second World-Wide Failure Exercise (WWFE-II) [56–58], and dedicated research by Lee et al. [59]. Despite originating from independent experimental projects, these data sources maintain high consistency in critical experimental protocols. The entire process, from specimen preparation to failure determination, is covered. Specifically, all tests utilize standard tubular specimens, with final fracture serving as the unified failure criterion. For complex loading conditions, standardized testing procedures are also followed, involving proportional loading until failure. The resulting dataset encompasses various common fiber-reinforced polymer FRP composite laminates and covers complex loading scenarios ranging from biaxial to triaxial. It is worth noting that for the thin-walled tubular specimens used in these benchmarks, the structural collapse often follows rapidly after the onset of critical damage mechanism. Therefore, in the context of the macroscopic evaluation adopted in this study, these experimental fracture points are utilized to define the engineering failure envelope.

A general schematic illustrating the multi-axial loading environment and the experimental configuration is depicted in Fig. 2. The central illustration details the standard tubular specimen geometry subjected to combined axial, transverse (induced by internal pressure), and torsional shear loads, along with typical laminate stacking sequences. To provide

a view of the dataset diversity, the surrounding visualization displays the proportional composition of stress components for all 16 distinct test cases. As shown, the dataset encompasses a wide spectrum of loading scenarios, ranging from dominant axial or shear states to complex coupled multi-axial conditions. Detailed quantitative information for each case, including specific failure strengths—which vary from under 500 MPa to over 2500 MPa—is tabulated in Table 1.

The biaxial loading data is primarily sourced from WWFE-I [53–55] and the work of Lee et al. [59]. This subset includes various material systems, such as E-glass/epoxy and T300 carbon/epoxy, accounting for a total of 330 failure data points. In contrast, the data for triaxial loading conditions originates solely from WWFE-II [56–59]. This portion of the dataset describes the failure behavior of five different orthotropic laminates, including T300/PR319, E-glass/MY750, S-glass/epoxy, A-S carbon/epoxy, and IM7/8551-7, each with a distinct fiber and matrix combination, under three-dimensional stress states.

2.2. Feature and label engineering for multi-axial failure analysis

A feature and label engineering process is necessary to convert the raw experimental data into a structured format suitable for the neural network model. The main objective of this process is to construct an input feature space that describes the material system, laminate architecture, and loading conditions. This process also defines a clear and low-dimensional prediction target.

2.2.1. Input feature construction

As illustrated in the Phase 2 of Fig. 3, the model utilizes a 20-dimensional feature vector as input. This feature space is constructed by aggregating three distinct physical information groups: Material properties (9D), Laminated architecture (5D), and Stress ratio (6D). Specifically, nine inputs define the basic strength properties of a single ply, including longitudinal, transverse, and shear strengths. An additional five inputs encode the structural information, covering the stacking

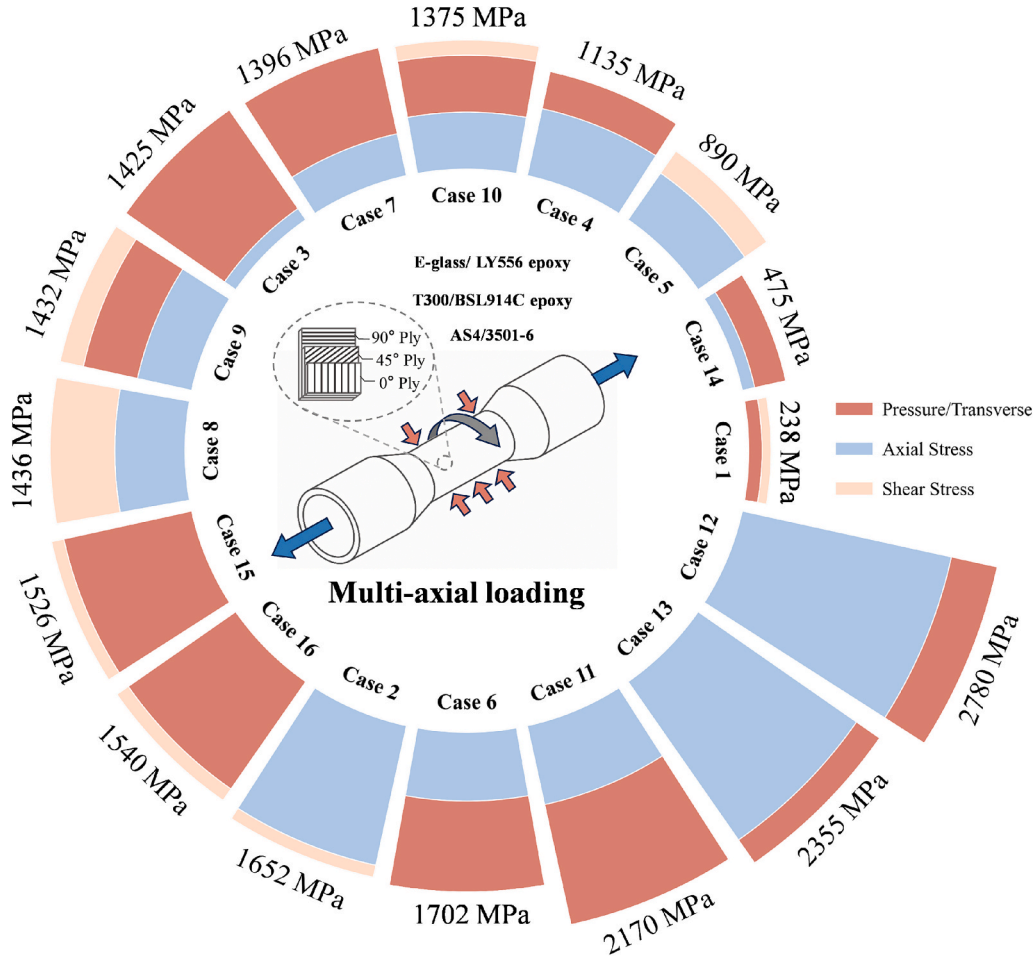


Fig. 2. Experimental configuration and stress state distribution of the multi-axial failure dataset. The central schematic illustrates the standard tubular specimen geometry subjected to combined axial, transverse (pressure), and torsional shear loading, detailing typical laminate stacking sequences. The surrounding visualization displays the proportional composition of stress components (Axial, Transverse, and Shear) for all 16 distinct test cases, while the outer numerical annotations indicate the maximum failure strength (in MPa) for each case, highlighting the diversity of loading conditions covered in the benchmark.

sequence and symmetry. Finally, to decouple the loading path from the stress magnitude, the complex multi-axial loading condition is normalized into six stress ratio components. This ensures the model learns the geometric direction of the load rather than its absolute value at the input level.

2.2.2. Target label definition

This framework simplifies the learning task through a decoupled prediction strategy. This approach differs from traditional data-driven methods that directly predict the high-dimensional stress tensor. The model's prediction target is defined as a single scalar output, L . This represents the magnitude of the failure stress vector. This value is calculated using the L^2 norm of the six stress components at the failure state, with the mathematical expression provided as follows:

$$L_{\text{true}} = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2 + \tau_{xy}^2 + \tau_{yz}^2 + \tau_{xz}^2} \quad (1)$$

The components σ_x, σ_y and σ_z represent the normal stresses, while τ_{xy}, τ_{yz} and τ_{xz} represent the corresponding shear stresses.

To preserve directional information while predicting only the scalar magnitude L , a six-dimensional unit direction vector $\mathbf{d}$ is independently calculated by normalizing the original failure stress tensor by L . This formulation allows the reconstruction of the stress tensor from the predicted scalar L during the training process. As visually summarized in Fig. 3, the workflow integrates these components to generate three core data matrices for model training: the 20-dimensional input feature

matrix (X_{train}), the 1-dimensional target label vector (L_{train}), and the 6-dimensional direction vector matrix.

2.2.3. Dual-normalization for data and physics domains

A dual-normalization strategy is implemented to process model inputs and evaluate physical constraints. First, min-max scaling independently transforms each of the 20 input features and the target label L into a normalized range of [0,1]:

$$v_{\text{scaled}} = \frac{v - v_{\min}}{v_{\max} - v_{\min}} \quad (2)$$

Here, $v_{\min}$ and $v_{\max}$ represent the minimum and maximum values of the variable observed across the training dataset.

For physics loss calculation, the model's scalar output L_{scaled} is first reverted to its physical magnitude L_{pred} and combined with the direction vector $\mathbf{d}$ to reconstruct the full stress tensor σ_{pred} :

$$\sigma_{\text{pred}} = L_{\text{pred}} \cdot \mathbf{d} \quad (3)$$

The evaluation of all physical criteria does not occur in the original physical stress space. This avoids the numerical challenges arising from large scale differences between stress components, such as longitudinal and shear stresses.

Subsequently, to establish a consistent evaluation domain, both the reconstructed stresses and material strength properties are mapped into a normalized physics space based on the global range of stress compo-

Table 1

Summary of the experimental datasets detailing for all 16 test cases.

Case	Material	Laminate	Loading	No. points
1	E-glass/ LY556 epoxy	0°	σ_y versus τ_{xy}	16
2	T300/BSL914C epoxy	0°	σ_x versus τ_{xy}	34
3	E-glass/MY750 epoxy	85°	σ_x versus σ_y	21
4	E-glass/LY556 epoxy	(90° / $\pm 30^\circ$ / 90°)	σ_x versus σ_y	47
5	E-glass/ LY556 epoxy	(90° / $\pm 30^\circ$ / 90°)	σ_x versus τ_{xy}	44
6	AS4/3501-6	(0° / $\pm 45^\circ$ / 90°)	σ_x versus σ_y	43
7	E-glass/MY750 epoxy	$\pm 55^\circ$	σ_x versus σ_y	82
8	T300/epoxy	(0° / 90°)	σ_x versus τ_{xy}	43
9	T300/PR319	0°	$\sigma_1 = \sigma_2 = \sigma_3$ versus τ_{12}	21
10	T300/PR319	90°	$\sigma_1 = \sigma_2 = \sigma_3$ versus τ_{12}	21
11	E-glass/MY750 epoxy	90°	σ_1 versus σ_3 ($\sigma_1 = \sigma_2$)	18
12	S-glass/epoxy	0°	σ_1 versus σ_3 ($\sigma_2 = \sigma_3$)	22
13	A-S carbon/epoxy	0°	σ_1 versus σ_3 ($\sigma_2 = \sigma_3$)	14
14	E-glass/MY750 epoxy	$\pm 35^\circ$	σ_y versus σ_z ($\sigma_x = \sigma_y$)	11
15	IM7/8551-7	(0° / 90° / $\pm 45^\circ$)	σ_z versus τ_{yz} ($\sigma_x = \sigma_y = 0$)	14
16	IM7/8551-7	(0° / 90°)	σ_z versus τ_{yz} ($\sigma_x = \sigma_y = 0$)	8

nents. Accordingly, material parameters such as the tensile strength X_T are normalized as:

$$X_{T_{\text{norm}}} = \frac{X_T}{(\sigma_{x_{\text{max}}} - \sigma_{x_{\text{min}}})} \quad (4)$$

This universal framework ensures that all subsequent physics-based loss calculations are performed within a stable and consistently scaled domain. It serves as the theoretical foundation for implementing the

three distinct PCNN models detailed in [section 3](#).

2.2.4. Feature correlation analysis

A feature correlation analysis is conducted to investigate the internal relationships within the constructed feature set. The calculation formula is as follows:

$$\rho = \frac{\text{cov}(\text{rg}_X, \text{rg}_Y)}{\sigma_{\text{rg}_X} \sigma_{\text{rg}_Y}} \quad (5)$$

Here, rg_X and rg_Y represent the ranked variables of the original datasets X and Y . The term $\text{cov}(\text{rg}_X, \text{rg}_Y)$ is the covariance of these ranked variables. This method is more robust to nonlinear dependencies and data outliers, which makes it suitable for the analysis of complex physical systems, particularly multiaxial loading scenarios governed by competing and complex damage mechanisms.

The analysis encompassed all 20 input features, which include 14 material and structural properties and 6 loading direction components, as well as the single target label, failure strength L . The results are visualized in the correlation matrix heatmap of [Fig. 4](#).

The heatmap reveals the relationships between features. First, strong positive correlations exist within the material strength properties, such as the 0.95 correlation coefficient between longitudinal tensile strength and longitudinal compressive strength. This reflects the intrinsic connection between a material's mechanical properties. In contrast, the correlation between different physically distinct feature groups—namely material properties, ply orientation parameters(θ_i), and loading direction components(d_i)—is generally weak. This indicates a low level of internal redundancy in the constructed feature set.

Most importantly, the analysis shows that no single input feature exhibits a dominant relationship with the final prediction target L . The absolute values of all correlation coefficients are at a low level. This lack of a simple monotonic relationship demonstrates that the failure strength of composite materials is a complex response determined by the nonlinear interaction of multiple physical factors. This complexity validates the effectiveness of the current dataset for developing a robust predictive model.

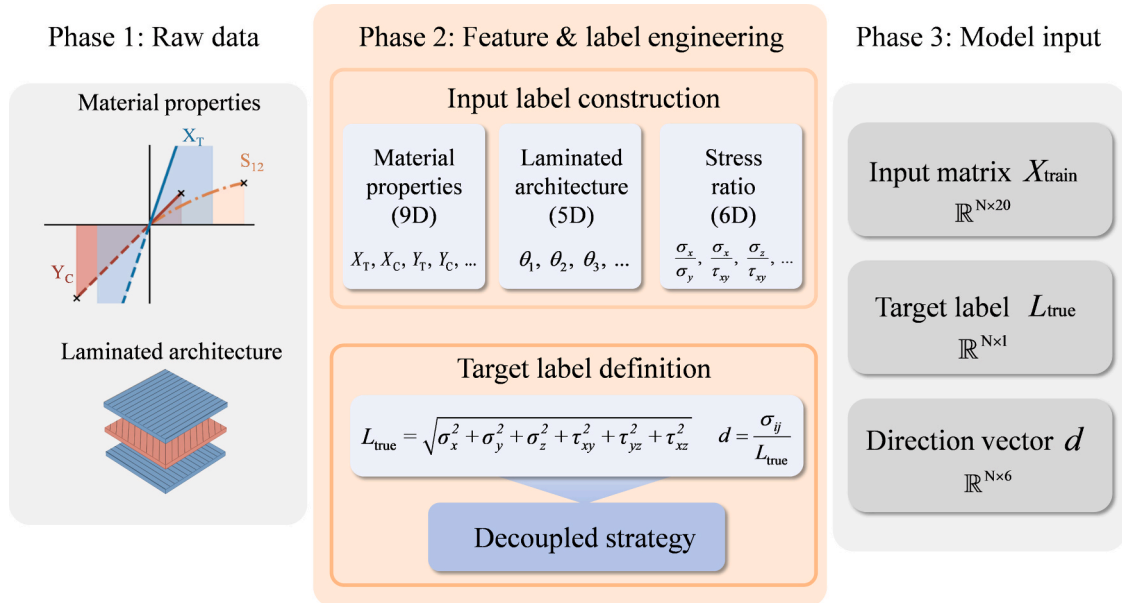


Fig. 3. Workflow of the feature engineering and data preprocessing pipeline. The schematic illustrates the multi-phase transformation of raw experimental records into structured training matrices. Key processes include the construction of the 20-dimensional input feature space encompassing material, structural, and loading parameters, and the target label definition utilizing the decoupled prediction strategy to decompose the stress tensor into a scalar failure magnitude and a normalized direction vector (d).

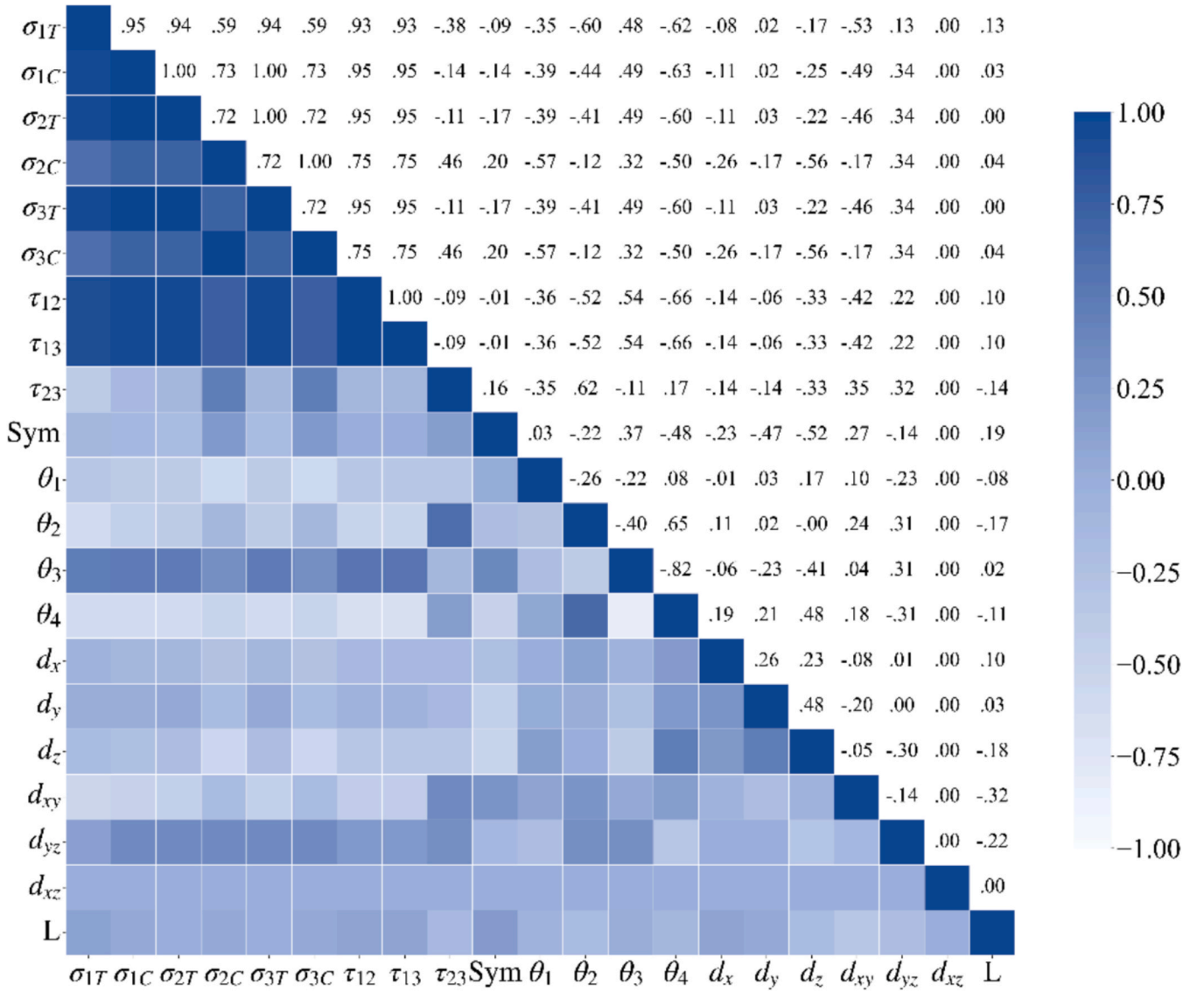


Fig. 4. Spearman correlation heatmap of the feature space. The distinct block structure highlights strong multicollinearity within material properties (top-left) and the independence of loading parameters. Notably, the consistently weak correlations suggest a non-linear mapping from inputs to L , supporting the use of non-linear regression models rather than linear regression.

3. Methodology

This section presents the framework for constructing and training a Physics-Constrained Neural Network, known as a PCNN. The concept is to integrate classical failure criteria as physical constraints into the neural network training process. This approach enhances the model's physical consistency and generalization capability while ensuring high accuracy in fitting experimental data. The overall technical workflow corresponds to the architecture visually summarized in Fig. 1. Input data is first processed and then fed into the PCNN model to predict a scalar failure strength, L . The model's training is guided by a hybrid loss function. The data loss term measures the deviation between the predicted values and the experimental values. The physics loss term is derived by reconstructing the predicted scalar L into a full stress tensor and then quantifying the extent to which this tensor violates the chosen failure criterion.

In what follows, section 3.1 introduces three classical failure criteria, namely Tsai-Wu [8], Cuntze [9], and LaRC [10–12], which provides the theoretical foundation for the physical constraints. Sections 3.2 and 3.3 detail the construction of a purely data-driven baseline model, an

Artificial Neural Network, and three distinct PCNN models with physical constraints. The network architectures and hybrid loss mechanisms are also described. Section 3.4 explains the hyperparameter optimization, nested cross-validation strategies, and performance evaluation metrics.

3.1. Multi-fidelity classical failure criteria

Predicting the strength of composite materials under complex stress states is a critical part of structural design. The purpose of a failure criterion is to define a material's safe operating domain within a multidimensional stress space. This domain is bounded by a failure envelope, which is illustrated in Fig. 5. Material failure is considered to occur when the stress state reaches or surpasses this envelope. Therefore, accurately describing the failure envelope is the central task of strength theory.

To incorporate diverse levels of physical inductive bias into the neural network, we select three representative classical failure criteria—Tsai-Wu, Cuntze, and LaRC, spanning from macroscopic phenomenological models to refined micromechanical theories. In this manuscript, 'fidelity' is used to describe the mechanistic richness of an

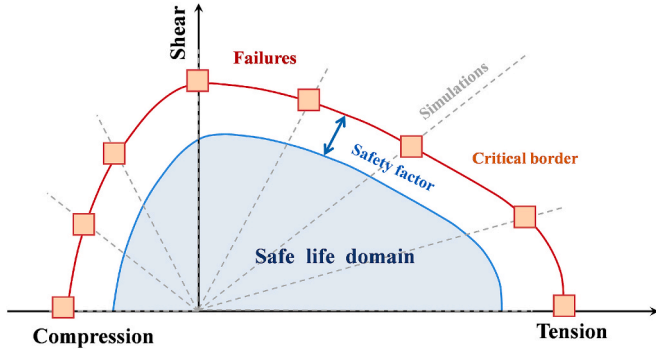


Fig. 5. Schematic definition of the failure envelope and safety margins in stress space. The geometric illustration delineates the Safe life domain (blue region) from the Critical border (red line), which represents the failure limit. Radial dashed lines indicate proportional loading paths used to quantify the Safety factor, defined as the reserve ratio between the operational stress state and the ultimate failure strength.

algebraic criterion within classical strength theories. This terminology does not imply that a laminate-scale progressive damage description is fully resolved by these criteria.

Tsai-Wu criterion serves as the representative macroscopic model, assuming a single, smooth, and closed failure surface described by a unified quadratic tensor polynomial. For a three-dimensional orthotropic material, the failure condition is expressed as:

$$F_i \sigma_i + F_{ij} \sigma_i \sigma_j = 1 \quad (i, j = 1, 2, \dots, 6) \quad (6)$$

Here, $\sigma_i = [\sigma_1, \sigma_2, \sigma_3, \tau_{23}, \tau_{13}, \tau_{12}]^T$. F_i and F_{ij} are the first-order and second-order strength tensor coefficients, respectively.

In contrast, the Cuntze criterion introduces a micromechanical perspective by conceptualizing failure as a competition between Fiber Failure (FF) and Inter-Fiber Failure (IFF). A distinctive feature is the IFF mode, which assumes failure occurs on a potential failure plane defined by an angle θ . The failure index is maximized with respect to this angle:

$$\left(\frac{\sigma_n}{R_n}\right)^2 + \left(\frac{\tau_l}{R_l}\right)^2 + \left(\frac{\tau_t}{R_t}\right)^2 = 1 \quad (7)$$

R_n is the transverse strength (Y_T for tension or Y_C for compression), R_l is the in-plane shear strength, and R_t is the out-of-plane shear strength.

Furthermore, the LaRC04 criterion represents high-fidelity physically-based theories developed by NASA. It incorporates refined damage mechanisms, such as fiber kinking and the influence of normal stress on matrix shear strength. Specifically, under transverse compression, the matrix failure index accounts for the frictional effect on the failure plane:

$$I_M = \left(\frac{\tau_T}{S_T - \eta_T \sigma_n}\right)^2 + \left(\frac{\tau_P}{S_P - \eta_P \sigma_n}\right)^2 = 1 \quad (8)$$

Here, S_T and S_L are the transverse and longitudinal (parallel to fibers) shear strengths, while η_T and η_L are friction coefficients for transverse and longitudinal, respectively. The detailed constitutive equations and parameter definitions for all three criteria are provided in [Appendix A](#).

It is therefore necessary to clarify the modeling level at which these criteria are used in the present study[60]. While LaRC04 [61] and Cuntze [9] criteria were originally formulated for unidirectional (UD) laminae, the benchmark database considered here reports laminate-level fracture points for multidirectional tubular laminates under proportional multiaxial loading (Section 2.1). For the thin-walled specimens in these benchmarks, global collapse commonly follows shortly after the governing damage mechanism becomes unstable; consequently, the measured fracture points provide a practical laminate-level limit state for constructing an engineering failure envelope. Consistent

with this benchmark definition, the laminate is represented at the envelope level as an equivalent orthotropic continuum, and the selected criteria are employed as algebraic references to regularize admissible envelope topology and mode competition, rather than as laminate-scale constitutive/damage laws. This envelope-level idealization (sometimes referred to as a simplified holistic approach in the WWFE context) intentionally abstracts away progressive stiffness degradation, inter-ply damage evolution (e.g., delamination), and non-proportional loading effects; these mechanisms are therefore treated as outside the scope of the present envelope reconstruction problem.

3.2. Establishing data-driven baselines

3.2.1. Deep learning baseline: The unconstrained ANN

To objectively evaluate the contribution of physical constraints, this study first employs a purely data-driven Artificial Neural Network as a baseline model. This model shares the same network architecture, input-output strategy, and hyperparameter search space as the subsequent PCNN models. This ensures a fair comparison of performance. The core difference is that the ANN relies on statistical patterns within the data for training and does not incorporate prior knowledge of physical failure criteria.

The baseline model employs a connected feed-forward neural network designed to execute the decoupled prediction strategy outlined in Section 2.2.2. By accepting a 20-dimensional feature vector as input, the network transforms the complex high-dimensional stress tensor prediction into a streamlined scalar regression task, outputting a single failure strength metric, L . Structurally, the network comprises multiple hidden layers equipped with advanced activation functions to capture the nonlinear dependencies within the feature space. The output layer utilizes a Sigmoid activation function to constrain the predicted values within the normalized interval (0,1). To ensure model robustness and mitigate overfitting, the architecture integrates Dropout regularization alongside specific weight initialization protocols.

In contrast to the physics-constrained models, this baseline is optimized in a purely data-driven manner, devoid of prior physical knowledge. The training objective is solely to minimize the Mean Squared Error (MSE) between the predicted failure strength L_{pred} and the true failure strength L_{true} :

$$Loss_{\text{data}} = \frac{1}{N} \sum_{i=1}^N (L_{\text{pred},i} - L_{\text{true},i})^2 \quad (9)$$

Here, N represents the number of samples in a training batch. The optimization process utilizes the Adam algorithm, augmented by an early stopping mechanism and dynamic learning rate scheduling to maximize generalization performance on unseen data.

3.2.2. Conventional machine learning benchmarks

To provide a performance evaluation against the current representative machine-learning baselines, and to isolate the specific contributions of the physical constraints, this study expands the comparative analysis by introducing six widely established machine learning algorithms as additional baselines. These models were carefully selected to cover a diverse spectrum of data-driven methodologies, ensuring an assessment of predictive capabilities across different learning paradigms.

To evaluate performance using classical non-parametric and kernel-based approaches, we incorporated K-Nearest Neighbors (KNN) and Support Vector Regression (SVR). KNN serves as a representative instance-based learner. For a given query input vector $\mathbf{x}$, the algorithm identifies the set $\mathcal{N}_k(\mathbf{x})$ containing the k nearest training samples based on the Euclidean distance metric $\|\mathbf{x} - \mathbf{x}_i\|_2$. The predicted failure strength is then computed as the local average of the neighbors' values:

$$\hat{y}_{\text{KNN}}(\mathbf{x}) = \frac{1}{k} \sum_{i \in \mathcal{I}_k(\mathbf{x})} y_i \quad (10)$$

Conversely, SVR addresses non-linearity by mapping the input space into a high-dimensional feature space via a mapping function $\phi(\mathbf{x})$. The regression function is formulated as:

$$f(\mathbf{x}) = \sum_{i=1}^N (\alpha_i - \alpha_i^*) K(\mathbf{x}_i, \mathbf{x}) + b \quad (11)$$

where α_i, α_i^* are Lagrange multipliers. By utilizing the kernel trick, SVR maps the input space into a high-dimensional feature space to find the optimal separating hyperplane, providing a reference for handling small-sample non-linear regression problems.

Given the prominence of tree-based ensembles in handling structured data, three representative machine-learning baselines were implemented: Random Forest (RF), XGBoost, and LightGBM. Random Forest employs a bagging strategy, averaging the predictions of multiple de-correlated decision trees to reduce variance. In contrast, XGBoost and LightGBM leverage a scalable gradient boosting framework, where trees are built sequentially to correct the errors of previous iterations. The prediction at step t is updated additively:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i) \quad (12)$$

Here, $f_t(\mathbf{x}_i)$ represents the new tree trained to fit the residuals of the previous step, and η is the learning rate. These algorithms currently represent a strong baseline for purely data-driven approaches in tabular regression tasks.

Additionally, a standard Deep Neural Network (DNN) was included to serve as a direct control group for the PCNN. This model shares a similar multilayer perceptron architecture with the proposed framework but is trained exclusively on data fidelity loss without any physical regularization. This inclusion allows us to isolate the specific impact of the physical constraints, verifying whether architectural depth alone is sufficient for high-fidelity failure envelope reconstruction.

To ensure a fair comparison, a systematic hyperparameter optimization was conducted for each baseline algorithm. We employed a grid search strategy on the validation set to identify the optimal configuration for critical governing parameters, such as the number of estimators and maximum depth for tree-based models, the kernel function and penalty parameter C for SVR, and the number of neighbors k for KNN. Crucially, all baseline models were trained and tested using the identical 20-dimensional input feature space and scalar target definitions as the PCNN framework, ensuring that any observed performance disparity is attributable solely to the modeling methodology rather than discrepancies in data processing.

3.3. Formulation of the physics-constrained neural network framework

Building upon the purely data-driven baseline model established in Section 3.2, this section formulates the core theoretical framework of the Physics-Constrained Neural Network. The essence of this formulation lies in a systematic methodology that integrates prior physical knowledge from classical failure criteria into the deep learning model's training process. Specifically, this integration is achieved by constructing a hybrid loss function that penalizes deviations from established physical laws. This approach ensures the model not only fits the observational data but also adheres to physical principles, thereby enhancing its physical consistency and generalization capability.

3.3.1. Core concept and hybrid loss function

To address the asynchronous magnitude changes between data and physics losses, we employ an adaptive weighting mechanism instead of a static weight. Governed by a loss target ratio α , this mechanism dynamically calculates the weighting coefficient at each training iteration

to ensure the physical loss magnitude aligns with the data loss according to the defined proportion. Consequently, the total loss function is constructed as:

$$Loss_{\text{total}} = Loss_{\text{data}} + \lambda_{\text{adaptive}} \cdot Loss_{\text{phys}} \quad (13)$$

The data loss term $Loss_{\text{data}}$ quantifies the fitting error utilizing the Mean Squared Error as defined in Equation (9). The physics loss term $Loss_{\text{phys}}$ measures the degree to which predictions violate the established failure criteria, with its specific formulation dependent on the physical theory employed. The weighting coefficient $\lambda_{\text{adaptive}}$ controls the strength of the physical constraint and serves as a critical hyperparameter for balancing data fitting with physical consistency.

In the proposed PCNN framework, these physical criteria are integrated as “soft” constraints within the loss function, rather than serving as absolute governing equations. We recognize that applying UD-based theories to laminates involves simplifications, particularly regarding in-situ strength effects and layer-wise constraints [61]. However, the experimental data of laminates inherently embodies these complex interaction mechanisms. Therefore, the PCNN employs a complementary strategy grounded in residual learning principles. Facilitated by the adaptive weighting mechanism ($\lambda_{\text{adaptive}}$), the framework dynamically balances theoretical guidance with observational reality. In this dynamic, the physical loss component guides the network to respect the geometric properties of the failure envelope (e.g., closure and convexity), while the data-driven component implicitly learns the constitutive residuals—specifically, the difference between the idealized theoretical prediction and the actual material response. This allows the model to utilize the analytical theory as a baseline, while automatically calibrating the predictions to account for the inter-ply mechanics that simplified theoretical models may overlook.

As visually detailed in Fig. 6, the essence of this hybrid loss mechanism is to constrain the solution space of the model. It restricts the solution from the set of all functions that can fit the data to a smaller subset of functions that both fit the data and comply with physical laws. The physics loss term plays the role of implicit regularization during training.

3.3.2. Stress tensor reconstruction mechanism

The calculation of the physics loss depends on the complete six-dimensional stress tensor, whereas the direct output of the neural network is only the scalar failure strength, L . Therefore, a stress tensor reconstruction mechanism is required to convert L into the stress state needed for the physics loss calculation. As described in Section 2.2, the reconstruction strategy employed in this study is based on the assumption of proportional loading. Specifically, the stress tensor at failure, σ^* , can be reconstructed using the following relationship:

$$\sigma^* = L \cdot \mathbf{r} \quad (14)$$

Here, $\mathbf{r} = [\sigma_x/\sigma_y, 1, \sigma_z/\sigma_y, \tau_{xy}/\sigma_y, \tau_{xz}/\sigma_y, \tau_{yz}/\sigma_y]^T$ is the normalized loading path vector, which is provided directly by the input features.

To ensure that the physical constraints are comparable across different material systems and loading conditions, all stress components and material strength parameters are normalized before the physics loss is computed. A normalized stress component, σ'_i , is defined as:

$$\sigma'_i = \frac{\sigma_i - \sigma_{\min,i}}{\sigma_{\text{range},i}} \quad (15)$$

In this normalized space, the criterion values for all failure criteria are scaled to a dimensionless form of order one. This ensures that the physics loss has a consistent physical meaning across different cases.

3.3.3. Constraint strategies: Equality and inequality

Based on the mathematical formulation and physical interpretation of the failure criteria, we further employ two different types of physical constraint strategies.

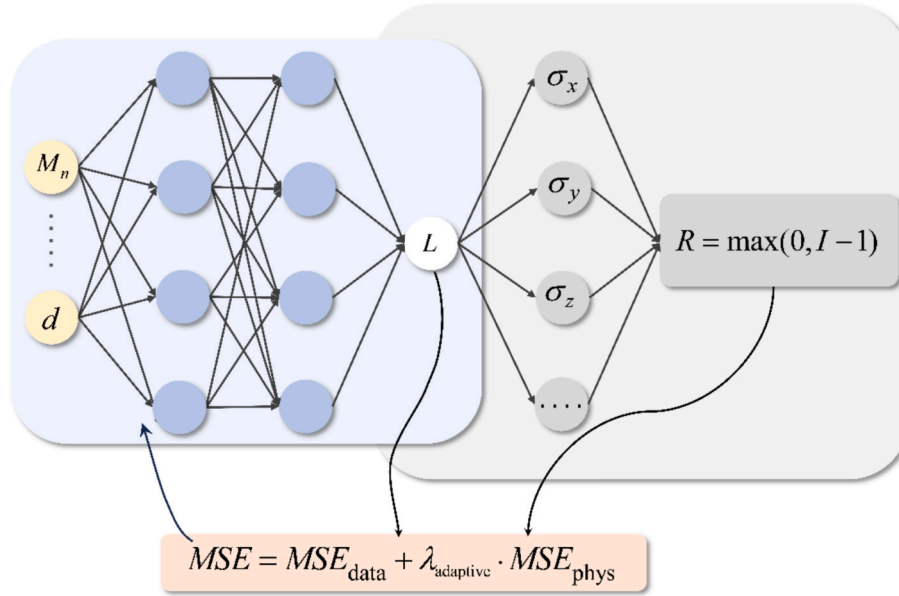


Fig. 6. Operational flow of the hybrid loss function with inequality constraints. The diagram details the stress reconstruction mechanism that bridges the data-driven scalar output L and the physics-based tensor evaluation. It illustrates the specific implementation of the inequality penalty strategy, where the physical loss acts as a regularization term to enforce the theoretical failure boundary.

The first is an equality constraint strategy, which is suitable for theories that express the failure criterion as a single, precise equation, such as the Tsai-Wu criterion. In this strategy, the predicted stress state is required to lie exactly on the failure envelope:

$$f(\sigma^*) = 1 \quad (16)$$

The corresponding physics loss is defined as the square of the theoretical residual.

$$Loss_{phys} = \frac{1}{N} \sum_{i=1}^N (f(\sigma_i^*) - 1)^2 \quad (17)$$

This type of constraint adheres to the mathematical definition of the criterion, imposing a strong constraint on the shape of the failure envelope. However, it requires that the criterion itself be capable of describing all failure modes.

The second is an inequality constraint strategy, which is applicable to failure criteria that distinguish between multiple independent and competing failure modes, such as the Cuntze and LaRC criteria. Under this strategy, each failure mode index, I_k , is required to not exceed one:

$$I_k(\sigma^*) \leq 1, \quad k = 1, 2, \dots, M \quad (18)$$

The corresponding physics loss is defined using the $\max(0, \cdot)$ function, which ensures that a penalty is applied only to predictions that fall outside the failure envelope. This one-sided penalty form is consistent with the physical reality of competing modes, where the most critical mode dictates failure, allowing the indices of non-dominant modes to be less than one. This approach is therefore more flexible and better suited for describing complex failure behaviors.

$$Loss_{phys} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^M \max(0, I_k(\sigma_i^*) - 1)^2 \quad (19)$$

These two constraint strategies represent different implementation paths. The equality constraint uses a two-sided penalty, penalizing any deviation from the target value, which reflects a requirement for exact failure conditions. In contrast, the inequality constraint uses a one-sided penalty, penalizing only violations of the safe domain. The former constraint is more stringent, while the latter offers greater flexibility.

3.4. PCNN model variants

Based on the unified theoretical framework in Section 3.3, we detail three PCNN model variants that enforce different physical constraints. These models share an identical network architecture, consistent with the ANN baseline from Section 3.2. Their key distinction lies solely in the definition of the physics loss term, which we design to incorporate specific prior knowledge. We summarize the general implementation architecture in Fig. 7 and detail the formulation of each variant in the following subsections.

3.4.1. PCNN-TW (equality constraint model)

The PCNN-TW model incorporates the Tsai-Wu failure criterion as its physical constraint. As a classic macroscopic phenomenological theory, the Tsai-Wu criterion describes the failure behavior for all stress states using a single, unified quadratic tensor polynomial, as detailed in Section 3.1.1. According to the Tsai-Wu theory, the failure state must satisfy:

$$F_{ij}\sigma'_i\sigma'_j + F_i\sigma'_i = 1 \quad (20)$$

Here, σ'_i represents the normalized stress components, while F_i and F_{ij} are the strength tensor coefficients recalculated based on the normalized strengths. Employing an equality constraint strategy, the physics loss for the PCNN-TW model is defined as the mean squared value of this theoretical residual:

$$Loss_{phys,TW} = \frac{1}{N} \sum_{i=1}^N [F_{ij}\sigma'_{i,k}\sigma'_{j,k} + F_i\sigma'_{i,k} - 1]^2 \quad (21)$$

This model uses an equality constraint, requiring the predictions to fall precisely on the Tsai-Wu failure surface. Its underlying physical assumption is the existence of a single, smooth, and closed failure envelope.

3.4.2. PCNN-Cuntze

The PCNN-Cuntze model is based on the Cuntze failure criterion, the core of which is to distinguish between two key modes: fiber failure and inter-fiber failure, as described in Section 3.1.2 and Appendix A.2. However, the inter-fiber failure criterion in the original Cuntze theory

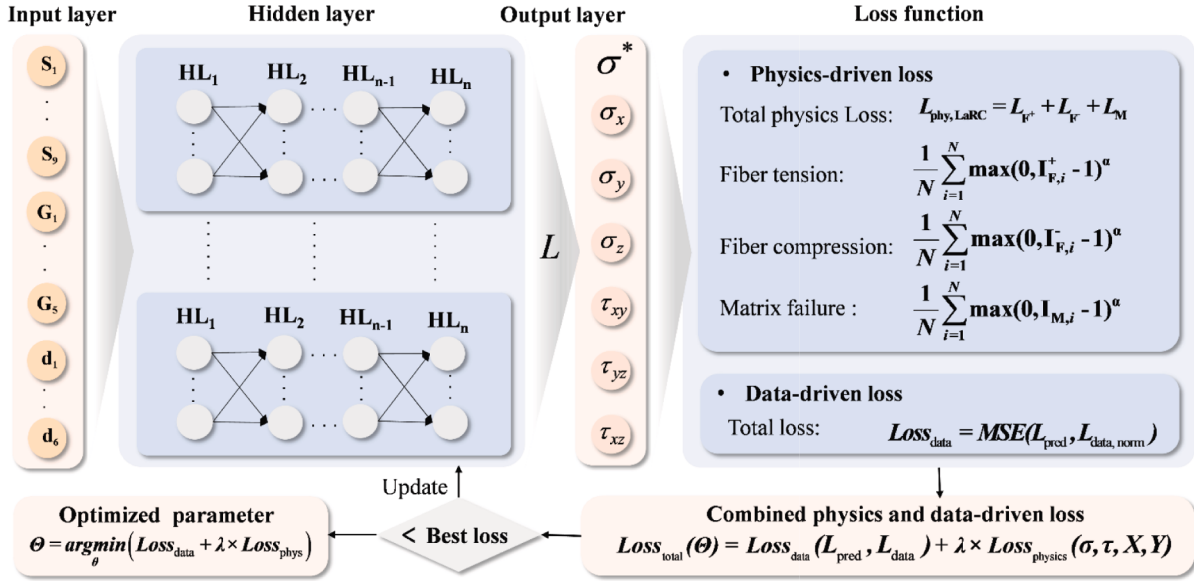


Fig. 7. Unified architecture of the Physics-Constrained Neural Network (PCNN) variants. The schematic illustrates the shared structural framework described in Section 3.4. It demonstrates the end-to-end flow from the input feature layer to the decoupled scalar output L . A critical component is the stress reconstruction module, which maps the predicted scalar back to the full stress tensor, guiding the network parameters θ toward a solution that satisfies both data fidelity and damage mechanisms.

involves an optimization search for the most dangerous fracture angle. This process is non-differentiable and cannot be directly embedded into the gradient-based training of a neural network.

To resolve this differentiability issue, we construct a Cuntze-inspired, decoupled physical model. It avoids the angle search and instead defines independent criteria for four primary failure modes:

Fiber Tension:

$$I_{\text{FT}} = \left(\frac{\sigma'_1}{X'_T} \right)^2 \quad (22)$$

Fiber Compression:

$$I_{\text{FC}} = \left(-\frac{\sigma'_1}{X'_C} \right)^2 \quad (23)$$

Matrix Normal Failure:

$$I_{\text{M}} = \left(\frac{\sigma'_2}{Y'_T} \right)^2 + \left(\frac{\sigma'_3}{Z'_T} \right)^2 \quad (24)$$

Shear Failure:

$$I_{\text{S}} = \left(\frac{\tau'_{12}}{S'_{12}} \right)^2 + \left(\frac{\tau'_{13}}{S'_{13}} \right)^2 + \left(\frac{\tau'_{23}}{S'_{23}} \right)^2 \quad (25)$$

Using an inequality constraint strategy, the model only penalizes violations of the safe domain where the failure index $I > 1$. The physics loss is defined as:

$$\text{Loss}_{\text{phys,Cuntze}} = \frac{1}{N} \sum_{i=1}^N \left[\max(0, I_{\text{FT},i} - 1)^2 + \max(0, I_{\text{FC},i} - 1)^2 + \max(0, I_{\text{M},i} - 1)^2 + \max(0, I_{\text{S},i} - 1)^2 \right] \quad (26)$$

Here, the $\max(0, \cdot)$ function implements a one-sided penalty, and the squared term applies a stronger penalty for more severe violations.

3.4.3. PCNN-LaRC (complete inequality model)

The PCNN-LaRC is the model variant with the highest physical fidelity in this study. It employs the LaRC series failure criterion, as detailed in Section 3.1.3 and Appendix A.3. The LaRC criterion not only

provides a detailed distinction between fiber tension, compression via micro-buckling, and matrix failure, but more critically, it offers an analytical expression for the most dangerous fracture angle. This feature circumvents the non-differentiable search problem.

The PCNN-LaRC model uses an inequality constraint. It defines the total physics loss as the sum of the losses from three independent modes:

$$\text{Loss}_{\text{phys,LaRC}} = \text{Loss}_{\text{F}+} + \text{Loss}_{\text{F}-} + \text{Loss}_{\text{M}} \quad (27)$$

For tensile fiber failure ($\sigma_x \geq 0$), the loss is defined as:

$$\text{Loss}_{\text{F}+} = \frac{1}{N_{\text{FT}}} \sum_{i \in \text{FT}} \max(0, I_{\text{F},i} - 1)^2 \quad (28)$$

The failure index for this mode is $I_{\text{F}} = \sigma'_x / X'_T$.

For compressive fiber failure ($\sigma_x < 0$), the loss is defined as:

$$\text{Loss}_{\text{F}-} = \frac{1}{N_{\text{FC}}} \sum_{i \in \text{FC}} \max(0, I_{\text{F},i} - 1)^2 \quad (29)$$

Here, the failure index I_{F} is calculated based on the micro-buckling model described in Section 3.1.3.

The matrix failure loss is defined as:

$$\text{Loss}_{\text{M}} = \frac{1}{N} \sum_{i=1}^N \max(0, I_{\text{M},i} - 1)^2 \quad (30)$$

The calculation of the matrix failure index I_{M} involves several steps. First, the fracture angle θ_0 is determined using an analytical formula. Then, the stresses are transformed to the failure plane coordinate system to calculate the normal stress σ'_n and the shear stresses τ'_T and τ'_L . Finally, the appropriate tensile or compressive criterion is selected based on the sign of σ'_n to evaluate the index.

3.5. Model training and evaluation

3.5.1. Network architecture and efficient training

Having detailed the theoretical frameworks for the baseline ANN and the three PCNN variants, we now describe the complete experimental workflow for their training, optimization, and evaluation. To ensure an

unbiased estimation of generalization performance and prevent information leakage during hyperparameter tuning, we employ a nested cross-validation strategy. This approach strictly separates model evaluation from hyperparameter optimization at the data level, as illustrated in Fig. 8.

A final hold-out test set is reserved prior to training. The remaining data undergoes a nested cross-validation process consisting of an outer loop for evaluation and an inner loop for hyperparameter tuning, both utilizing k -fold splits. Within each outer iteration, the inner loop identifies the optimal hyperparameters, which are subsequently applied to train the final model on the outer training set for validation.

An optimization was conducted covering network architecture, training strategies, and physical constraint mechanisms. The specific search space defined for each parameter, along with the identified optimal configuration employed in the final model, are detailed in Table 2.

As detailed in Table 2, the hyperparameter optimization converged to a compact network architecture defined by the configuration [20,512,1]. This notation denotes a streamlined Multi-Layer Perceptron (MLP) structure comprising a 20-dimensional input layer, a single wide hidden layer with 512 neurons, and a scalar output layer, rather than a deep, over-parameterized network. To maximize the non-linear representational capacity within this shallow depth, we employed the Swish-Gated Linear Unit (SwiGLU) as the activation function. SwiGLU enhances expressivity by introducing a gating mechanism that dynamically modulates feature information. Mathematically, for an input vector $\mathbf{x}$, the SwiGLU operation is defined as:

$$\text{SwiGLU}(\mathbf{x}, \mathbf{W}, \mathbf{V}, \mathbf{b}, \mathbf{c}) = \text{Swish}_\beta(\mathbf{x}\mathbf{W} + \mathbf{b}) \otimes (\mathbf{x}\mathbf{V} + \mathbf{c}) \quad (31)$$

where $\otimes$ denotes the element-wise product, and $\mathbf{W}, \mathbf{V}, \mathbf{b}, \mathbf{c}$ represents the weight matrices and bias vectors for the two parallel linear transformations (gate and value pathways), respectively. The Swish function is given by $\text{Swish}_\beta(\mathbf{z}) = \mathbf{z} \cdot \sigma(\beta\mathbf{z})$, where σ is the sigmoid function. This mechanism allows the network to capture complex coupling effects within the failure criteria without relying on excessive depth.

A complexity analysis further confirms the model's efficiency. The SwiGLU layer involves two parallel linear projections, resulting in 21,504 trainable parameters for the transformation from the 20-dimensional input to the 512-neuron hidden layer. The subsequent projection to the single output neuron contributes an additional 513 parameters. Consequently, the total model complexity is constrained to 22,017 parameters ($\approx 2.2 \times 10^4$). This lightweight architecture mitigates the risk of overfitting, a common challenge in data-driven composite analysis. Furthermore, the training process is computationally efficient; the final model was trained on a standard workstation equipped with an Intel Core i9-13900HX CPU, requiring no specialized GPU acceleration. This

Table 2

Definition of the hyperparameter search space and optimized configurations.

Category	Hyperparameter	Search Space	Optimal Value
Network	Architecture (Input, Hidden, Output)	Depth: 1–4; Width: [8,512]	[1,20,512]
	Activation function	[tanh, SwiGLU, ReLU]	SwiGLU
Training	Learning rate	$[10^{-4}, 2 \times 10^{-2}]$	0.01
	Dropout rate	[0.0, 0.2]	0.1
	Weight decay	[0.0, 0.01]	0.0
	Warm-up epochs	[0, 1000]	800
Physics	Physics weight (λ)	[0.0, 1.0]	Adaptive
	Loss target ratio (α)	[0.05, 0.3]	0.1

demonstrates the framework's suitability for rapid engineering deployment.

3.5.2. Uncertainty quantification via Monte Carlo dropout

Standard neural networks typically provide deterministic point predictions, which can be risky in safety composite design, particularly in regions with sparse experimental data. To quantify predictive uncertainty and construct reliable failure envelopes, this study employs Monte Carlo (MC) Dropout as a Bayesian approximation, following the theoretical framework established by Gal and Ghahramani. Unlike traditional training-only dropout, we maintain the dropout layers active (with a rate of $p = 0.1$ as optimized in Table 2) during the inference phase. For each input stress state $\mathbf{x}$, the model executes $N = 100$ stochastic forward passes, generating a probabilistic distribution of predicted failure strengths $\{L_1, L_2, \dots, L_T\}$.

Under the assumption that the predictive distribution can be approximated as Gaussian, the Mean Prediction (μ_{pred}) and the Sample Standard Deviation (σ_{pred}) are calculated to represent the macroscopic engineering failure limit and the associated predictive uncertainty, respectively:

$$\mu_{\text{pred}} \approx \frac{1}{T} \sum_{t=1}^T L_t(\mathbf{x}) \quad (32)$$

$$\sigma_{\text{pred}} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (L_t(\mathbf{x}) - \mu_{\text{pred}})^2} \quad (33)$$

In these formulations, T denotes the total number of Monte Carlo sampling iterations, set to 100 in this study to ensure statistical significance. The term $L_t(\mathbf{x})$ represents the predicted failure strength obtained in the t -th stochastic forward pass, where the active dropout mechanism ensures that each prediction corresponds to the output of a distinct sub-

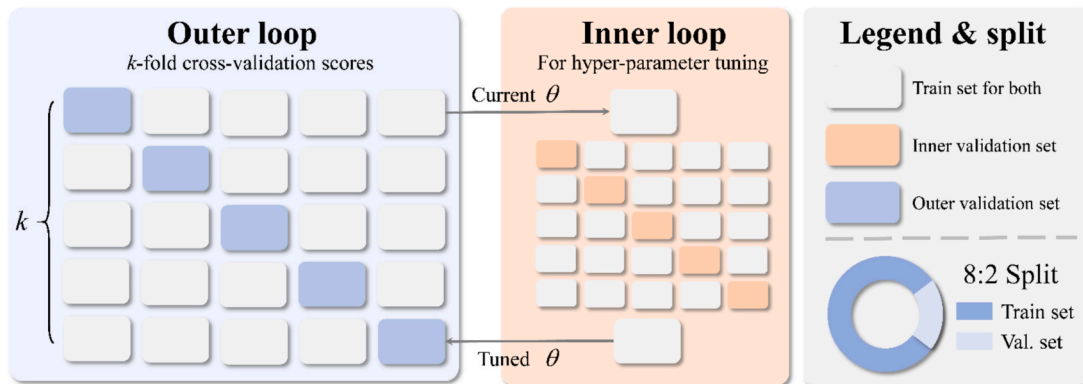


Fig. 8. Schematic of the nested cross-validation strategy employed for unbiased performance estimation. Visualization of the hierarchical data partitioning that isolates the Inner Loop (hyperparameter tuning) from the Outer Loop (model evaluation). This dual-loop structure operates independently of the final hold-out test set described in Section 3.5.

network configuration. The parameter μ_{pred} serves as the model's expected estimation of the failure limit, while σ_{pred} utilizes Bessel's correction (using $T - 1$) to provide an unbiased estimate of the dispersion. Physically, σ_{pred} primarily captures the epistemic uncertainty of the model; a larger standard deviation implies that the network parameters are less constrained by training data in that specific local region, resulting in divergent predictions.

To ensure a safety-oriented interpretation at the envelope level, we define the Conservative Design Envelope ($L_{\text{conservative}}$) based on the lower bound of the 95% Confidence Interval. Under the decoupled formulation adopted in this study, predicting L for a given loading direction d corresponds to locating the envelope intersection along the same proportional loading path. Accordingly, selecting the lower-tail estimate implies a smaller predicted failure magnitude on that path, i. e., an earlier attainment of the envelope boundary, and is therefore conservative in the present engineering sense. To adhere to physical constraints, a non-negativity condition is enforced:

$$L_{\text{conservative}} = \max(0, \mu_{\text{pred}} - 1.96 \cdot \sigma_{\text{pred}}) \quad (34)$$

This lower bound should therefore be interpreted as a probabilistic safety margin for envelope-level assessment and certification-oriented screening, rather than as a universal deterministic knock-down factor applicable to all design scenarios. Within this scope, the probabilistic strategy transforms the neural network from a 'black-box' regressor into a reliability-aware predictor.

3.5.3. Performance evaluation metrics

To evaluate the predictive performance of each model, we employed a set of standard evaluation metrics covering multiple dimensions. The Coefficient of Determination (R^2), Mean Relative Error (MRE), and

Normalized Root Mean Squared Error (NRMSE) are defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^N (L_{\text{true},i} - L_{\text{pred},i})^2}{\sum_{i=1}^N (L_{\text{true},i} - \bar{L}_{\text{true}})^2} \quad (35)$$

$$\text{MRE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{L_{\text{pred},i} - L_{\text{true},i}}{L_{\text{true},i}} \right| \times 100\% \quad (36)$$

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (L_{\text{pred},i} - L_{\text{true},i})^2}}{\max(L_{\text{true}}) - \min(L_{\text{true}})} \quad (37)$$

Here, $L_{\text{true},i}$ and $L_{\text{pred},i}$ are the true and predicted failure strengths for the i -th sample, $\bar{L}_{\text{true}}$ is the mean of all true failure strengths, and N is the total number of samples. This metric eliminates the influence of varying failure strength ranges on the error assessment, providing a fair basis for performance comparison across different cases. In subsequent sections, RMSE is referred to as this Normalized Root Mean Squared Error.

4. Results and discussion

4.1. Failure strength prediction: PCNN model performance

4.1.1. Overall evaluation of prediction fidelity

The overall prediction performance of PCNN models under three different physical constraint strategies (PCNN-TW, PCNN-Cuntze, and PCNN-LaRC) is displayed in Fig. 9. The accuracy of the predictions is evaluated by comparing the predicted scalar failure strength L_{pred} with experimental observations L_{true} .

Quantitative indicators demonstrate that three models achieve high

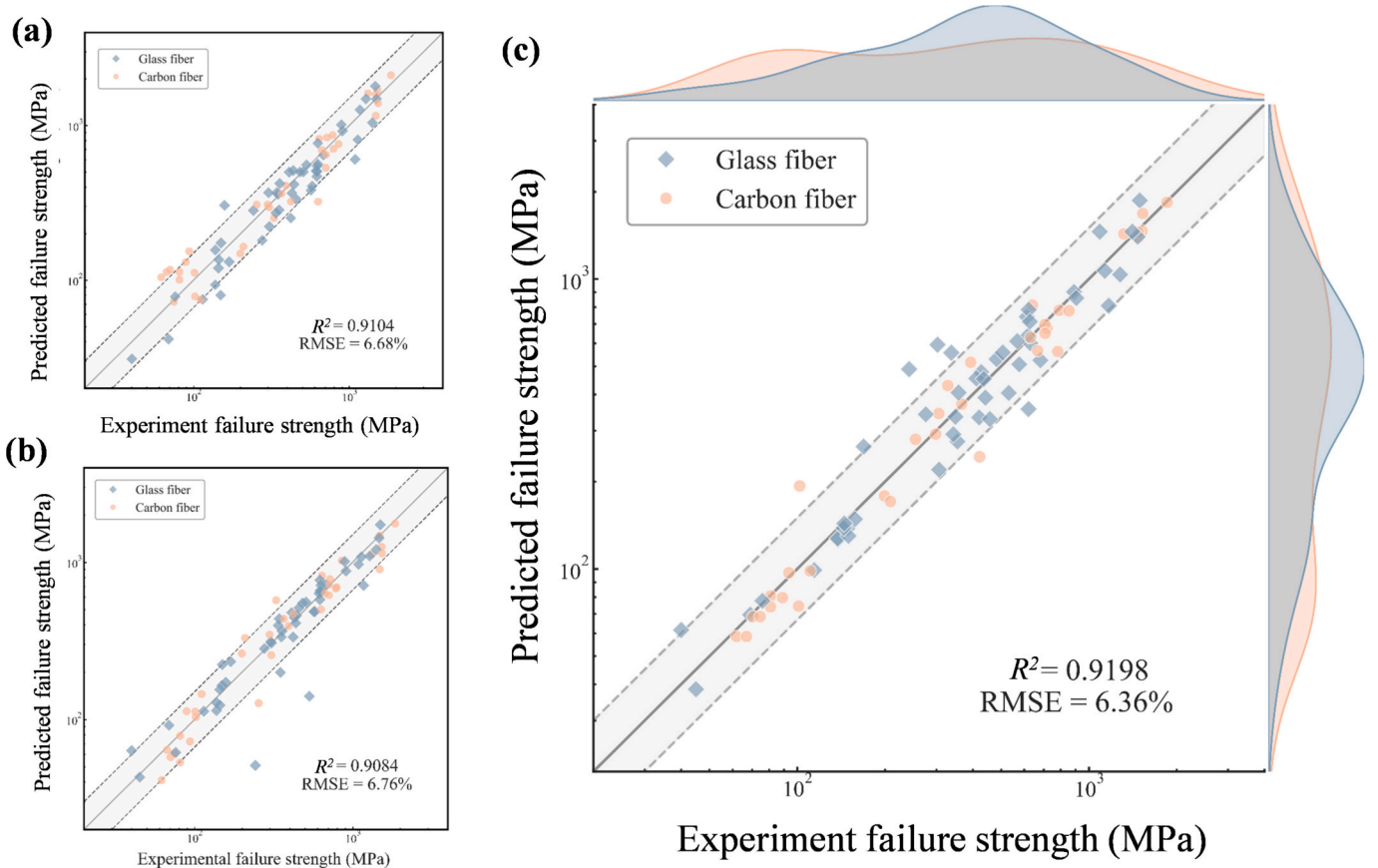


Fig. 9. Comparative evaluation of prediction fidelity across distinct physical constraint strategies. The scatter plots illustrate the correlation between predicted and experimental failure strengths for (a) PCNN-TW (Equality Constraint), (b) PCNN-Cuntze (Simplified Inequality), and (c) PCNN-LaRC (High-fidelity Inequality).

prediction accuracy. These results underscore the effectiveness of the physics-constrained framework. However, discernible differences in the metrics establish a distinct performance hierarchy across the models. The PCNN-Cuntze model based on simplified decoupled criteria achieves a baseline performance with an R^2 of 0.9084 as shown in Fig. 9b. In comparison, the PCNN-TW model using macroscopic phenomenological theory demonstrates prediction capability. Its R^2 increases to 0.9104 as shown in Fig. 9a. The PCNN-LaRC model based on high-fidelity damage mechanisms achieves the best performance among the three. It obtains the highest coefficient of determination of 0.9198 as shown in Fig. 9c.

These performance differences stem from variations in the physical fidelity of the constraints embedded within each model. The relatively lower performance of PCNN-Cuntze arises because its decoupled criteria ignore complex coupling effects among stress components. While it distinguishes main failure modes, this simplification limits its prediction capability under multiaxial stress states. The performance improvement of PCNN-TW benefits from the tensor polynomial form of the Tsai-Wu criterion. This form mathematically captures interactions between stress components through the interaction term F_{ij} . The PCNN-LaRC achieves the best results because its constraints originate from a refined characterization of real damage physical mechanisms. The LaRC criterion independently discriminates between fiber and matrix failure. More importantly, it accurately simulates physical details such as the influence of normal stress on shear strength on the potential failure plane. This physical prior knowledge not only provides the effective guidance for neural network optimization but also enables the model to learn data-driven solutions that are closer to physical reality—an advantage particularly prominent in complex stress regions with competing damage mechanisms.

4.1.2. In-depth performance analysis: Case studies

Although the overall performance is satisfactory, a more refined case-by-case analysis is crucial for deeply understanding model adaptability. This analysis covers different material systems, laminate structures, and loading conditions. Fig. 10 delivers key visual evidence for this analysis using the PCNN-LaRC model, where training and testing data points are distinguished in blue and red, respectively. For a concise quantitative comparison across all cases, the case-wise metrics are summarized in Table 3.

To complement the visual evidence in Fig. 10, Table 3 reports the

Table 3

Case-wise prediction performance of the PCNN-LaRC model across 16 test cases.

Case	N	R^2	NRMSE (%)	PWB
Case 1	16	0.5631	21.29	0.88
Case 2	34	0.9645	6.37	0.94
Case 3	21	0.382	23.18	0.67
Case 4	45	0.9342	7.01	1
Case 5	42	0.6087	13.56	0.9
Case 6	43	0.9503	5.43	0.77
Case 7	82	0.6496	17.69	0.74
Case 8	43	0.8855	12.32	0.77
Case 9	21	0.8482	13.1	0.71
Case 10	21	0.9759	5.01	1
Case 11	18	0.9583	6.58	0.78
Case 12	22	0.6906	15.36	1
Case 13	14	0.7783	15.83	1
Case 14	11	0.9071	10.94	0.73
Case 15	14	0.999	0.87	0.96
Case 16	8	0.9828	4.43	1

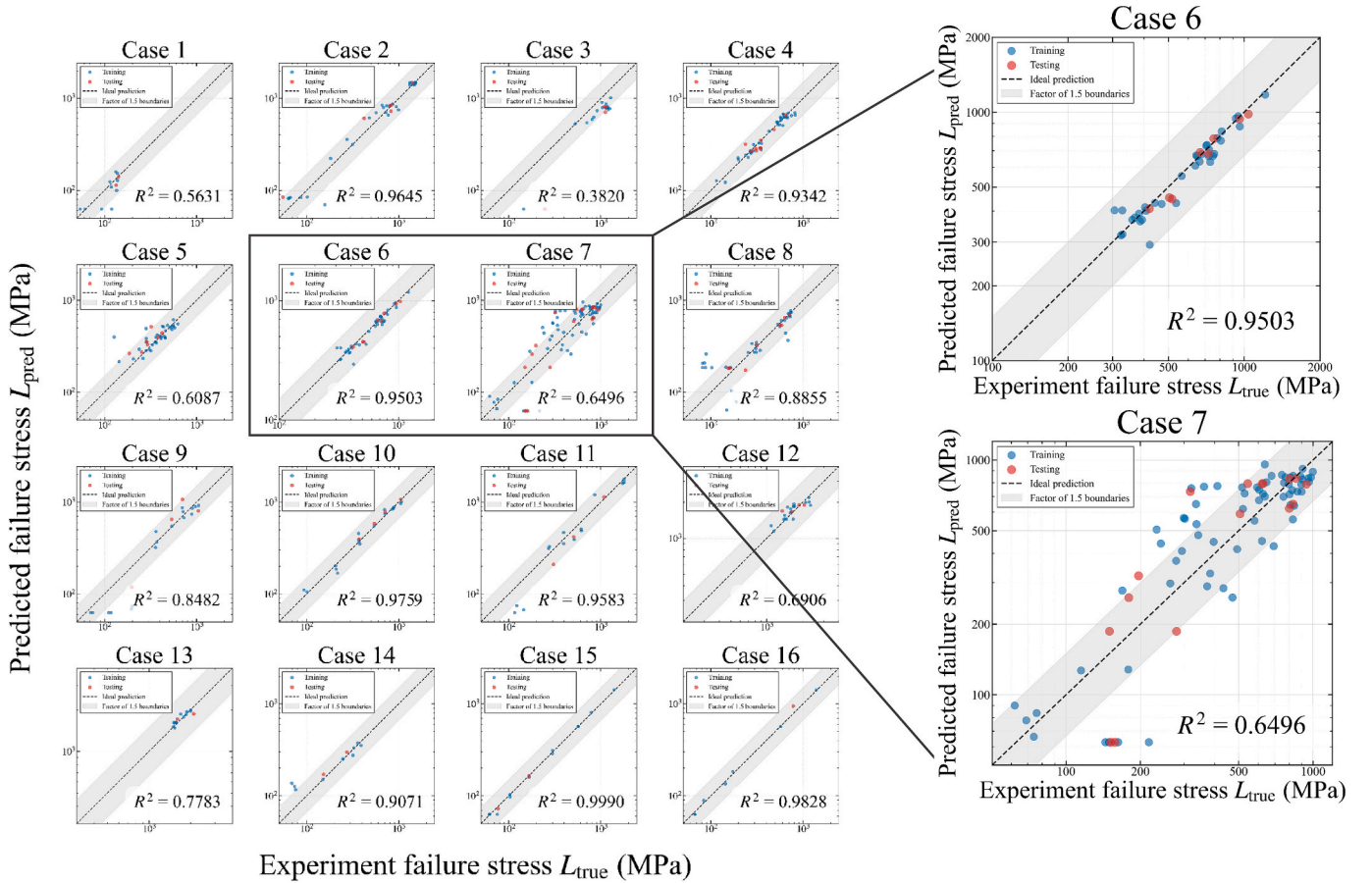


Fig. 10. Prediction performance across distinct material systems and loading conditions. Parity plots comparing experimental versus predicted failure strength for all 16 test cases, distinguishing between training (blue) and testing (red) subsets. Zoomed-in views illustrate the model's prediction capabilities in Case 6, while highlighting the challenges posed by pronounced experimental data dispersion in Case 7.

case-wise R^2 and the NRMSE normalized by the experimental strength range of each case. In addition, the Percentage Within Bounds (PWB) is included as an engineering-oriented indicator, defined as the fraction of predictions falling within a standard $1.5 \times$ tolerance envelope (i. e., $y/1.5 \leq \hat{y} \leq 1.5y$). This metric will be further discussed in the subsequent comparison against classical criteria.

A universal observation is that the model predictions on the test set exhibit high consistency with the training set for almost all cases. As shown in Fig. 10, the red testing points and blue training points jointly form a dense band along the ideal diagonal, with no systematic deviation or increased dispersion in the test set. This close agreement demonstrates the PCNN framework's generalization capability and its success in avoiding the overfitting typical in complex high-dimensional feature spaces—a direct result of the intrinsic advantage endowed by the physical constraints. The analysis shows that the PCNN model demonstrates reliable prediction capability in the vast majority of cases, and is particularly accurate for advanced carbon fiber composites. For instance, Case 6 (AS4/3501-6) and Case 2 (T300/BSL914C) represent typical aerospace-grade carbon fiber and epoxy resin systems. As summarized in Table 3, their R^2 values reach 0.9503 and 0.9645 respectively under in-plane biaxial tension and tension-shear coupling loading. Even more notable is that the model achieves an R^2 value close to a 0.9990 in Case 15 (IM7/8551-7). This case involves a complex three-dimensional stress state (σ_z vs τ_{yz}), demonstrating that the PCNN framework not only handles traditional plane stress problems but also effectively captures failure behaviors under such complex conditions—thereby proving its value for complexly loaded composite structure design.

However, a discernible divergence in predictive metrics is observed in E-glass systems, specifically in Case 1 (0°), Case 3 (85°), and Case 7 ($\pm 55^\circ$), where R^2 values stabilize in the range of 0.38 to 0.65. This performance variance should be attributed to the inherent stochasticity of the material response rather than model inconsistency. Unlike carbon fibers, E-glass composites often exhibit higher microstructural heterogeneity, including variations in fiber diameter and interfacial bonding strength. In Case 1 (longitudinal loading), sensitivity to initial fiber waviness and misalignment often leads to scatter in compressive strength. Similarly, Case 3 (transverse loading) and Case 7 (off-axis loading) are dominated by matrix and interface properties, where the failure initiation is notoriously sensitive to random distribution of micro-voids. These mechanisms introduce substantial aleatoric uncertainty into the experimental data, creating a “noise floor” that limits the theoretical maximum of any deterministic prediction model.

To objectively evaluate the model's utility in these high-dispersion

regimes, we conducted a comparative study against the classical LaRC04 and Tsai-Wu criteria, as visualized in Fig. 11. From a practical design perspective, the Percentage Within Bounds (PWB) metric serves to quantify usability. As reported in Table 3, this metric reflects the proportion of predictions falling within a standard $1.5 \times$ design tolerance envelope. The analysis reveals that while classical criteria provide valuable baselines, they struggle to accommodate the scatter observed in glass fiber systems. As shown in the right panel of Fig. 11, for the highly dispersed Case 7, the traditional LaRC04 criterion achieves a PWB of 0.51, indicating that nearly half of the experimental points fall outside the standard safety margins. In contrast, the PCNN framework enhances engineering reliability by adapting to the specific data distribution, elevating the PWB to 0.74. This advantage is even more pronounced in Case 3, where the LaRC04 criterion yields a PWB of only 0.15 due to the complex stress interactions, whereas the PCNN achieves a substantial improvement with a PWB of 0.77. This comparison suggests that while traditional criteria offer a solid starting point, the PCNN framework expands the reliable design space by accommodating experimental realities that rigid analytical formulations may overlook. This robustness is further underscored by the statistical fidelity analysis (Fig. 11, left panel). In the high-scatter environments of Case 3 and Case 7, the rigid manifolds of analytical theories fail to conform to the stochastic strength jumps, resulting in negative R^2 values (e.g., -8.22 for LaRC04 in Case 3 and -6.63 in Case 7). Conversely, the PCNN maintains positive correlations ($R^2 \approx 0.38$ for Case 3 and 0.65 for Case 7). This contrast demonstrates that the physically constrained network acts as a robust filter, successfully extracting the constitutive trend from the intrinsic noise without overfitting to outliers, thereby offering a consistent safety boundary that complements traditional analytical methods.

4.2. Expanded benchmarking and ablation analysis

To evaluate the efficacy of the proposed PCNN framework, we adopt a two-tiered validation strategy. First, we benchmark the model against a diverse set of representative machine-learning baselines to establish its relative performance standing. Second, we conduct a focused ablation study by comparing the PCNN with an architecturally identical but unconstrained Artificial Neural Network (ANN). This systematic approach allows us to isolate the specific contributions of physical constraints and verify whether the performance gains stem from the inductive bias rather than the learning capacity of the neural network itself.

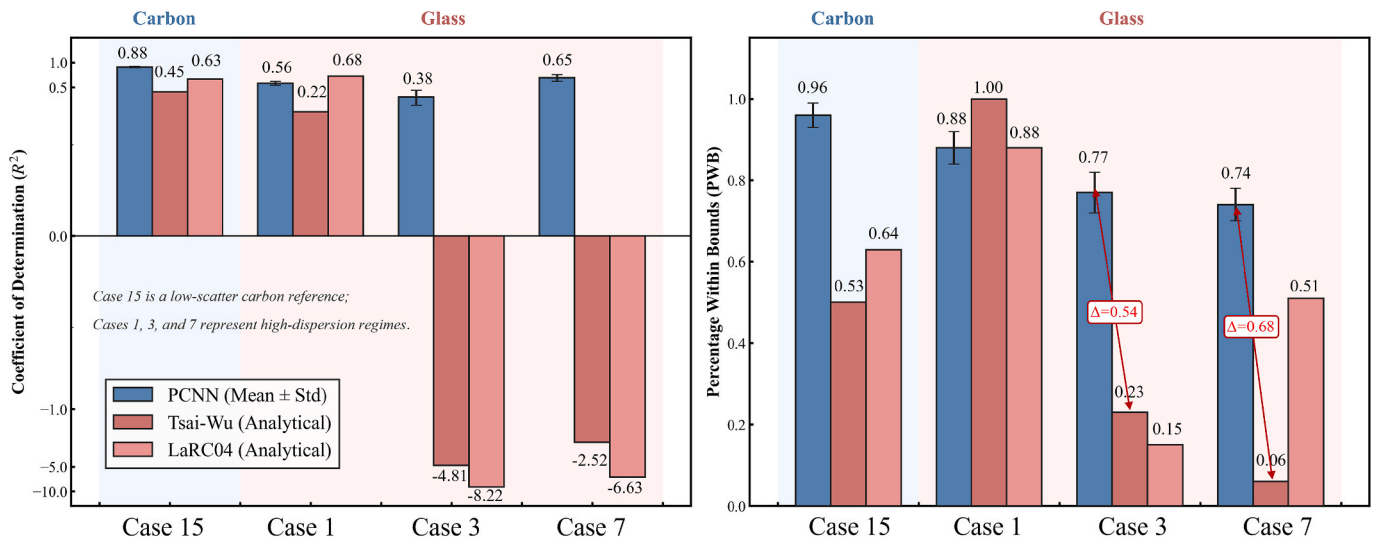


Fig. 11. Dual-perspective performance evaluation: Engineering utility versus statistical fidelity. The left panel compares the Coefficient of Determination across representative carbon (Case 15) and glass fiber cases (Case 1, 3, 7). The right panel contrasts the Percentage Within Bounds (PWB) metric.

4.2.1. Benchmarking against representative machine-learning baselines

To position the proposed framework within the broader landscape of data-driven mechanics, we benchmarked the PCNN-LaRC model against six established algorithms spanning instance-based learning (KNN), kernel methods (SVR), ensemble strategies (Random Forest, XGBoost, LightGBM), and standard deep learning (DNN). As illustrated in Fig. 12 (a–f), the comparative analysis reveals a clear performance hierarchy among the purely data-driven baselines. Tree-based ensemble methods emerged as the most robust approach for this tabular dataset, with XGBoost in Fig. 12b establishing a best observed baseline performance of $R^2 = 0.8769$, closely trailed by Random Forest and LightGBM. In contrast, the unconstrained Standard DNN yielded a lower determination coefficient of 0.8347 and conventional regressors such as SVR (Fig. 12a) exhibited limitations with an R^2 of 0.6506, struggling to capture the highly non-linear failure boundaries as effectively as the boosting algorithms. This disparity underscores that, without domain-specific guidance, increasing model complexity alone does not improve generalization in small-sample material regimes.

Compared with the best-performing baseline, the PCNN-LaRC framework demonstrates an advantage, attaining a Coefficient of Determination of 0.9198. While the best-performing ensemble models plateaued around an accuracy of 88%, the integration of physical constraints enabled our model to surpass this limit, reducing the prediction error compared to the strongest baseline. This result quantitatively demonstrates that the integration of physical constraints effectively regularizes the learning process, enabling the network to capture non-linear failure behaviors that are challenging for purely data-driven models. Consequently, the PCNN not only excels in numerical precision but also supports the usefulness of hybrid physics–data modeling

for this task.

4.2.2. Ablation study: The impact of physical constraints

To isolate the specific contribution of physical constraints and decouple the performance gains from the intrinsic learning capacity of the neural architecture, this section implements an ablation study. We systematically contrast the proposed PCNN framework against an architecturally identical but unconstrained ANN baseline. The evaluation strategy is structured across three progressive dimensions of difficulty: first, we assess the statistical fidelity and error distribution characteristics on the full test set; second, we examine the model stability under data-sparse regimes (ranging from 10% to 50% data availability); and finally, we challenge the generalization capability to unseen material systems via a Leave-One-Case-Out (LOCO) protocol. This hierarchical validation framework aims to quantitatively verify the hypothesis that physical laws serve as an inductive bias, effectively regularizing the black-box model and mitigating overfitting risks in complex mechanical regimes.

The prediction performance of this unconstrained baseline is visualized in Fig. 13a. Quantitatively, the model demonstrates considerable non-linear fitting capability, achieving an R^2 of 0.8660 and an NRMSE of 8.96%. These results confirm that a sufficiently deep feedforward network can learn the complex mapping from a high-dimensional feature space to material failure strength, capturing the main data trends. However, compared to the PCNN, its predictions exhibit more pronounced limitations, as evidenced by a higher dispersion of data points in Fig. 13. This is particularly evident in the medium-to-high strength range where $L > 1000$ MPa. Multiple prediction points deviate from the ideal diagonal line in this region. More critically, 12%

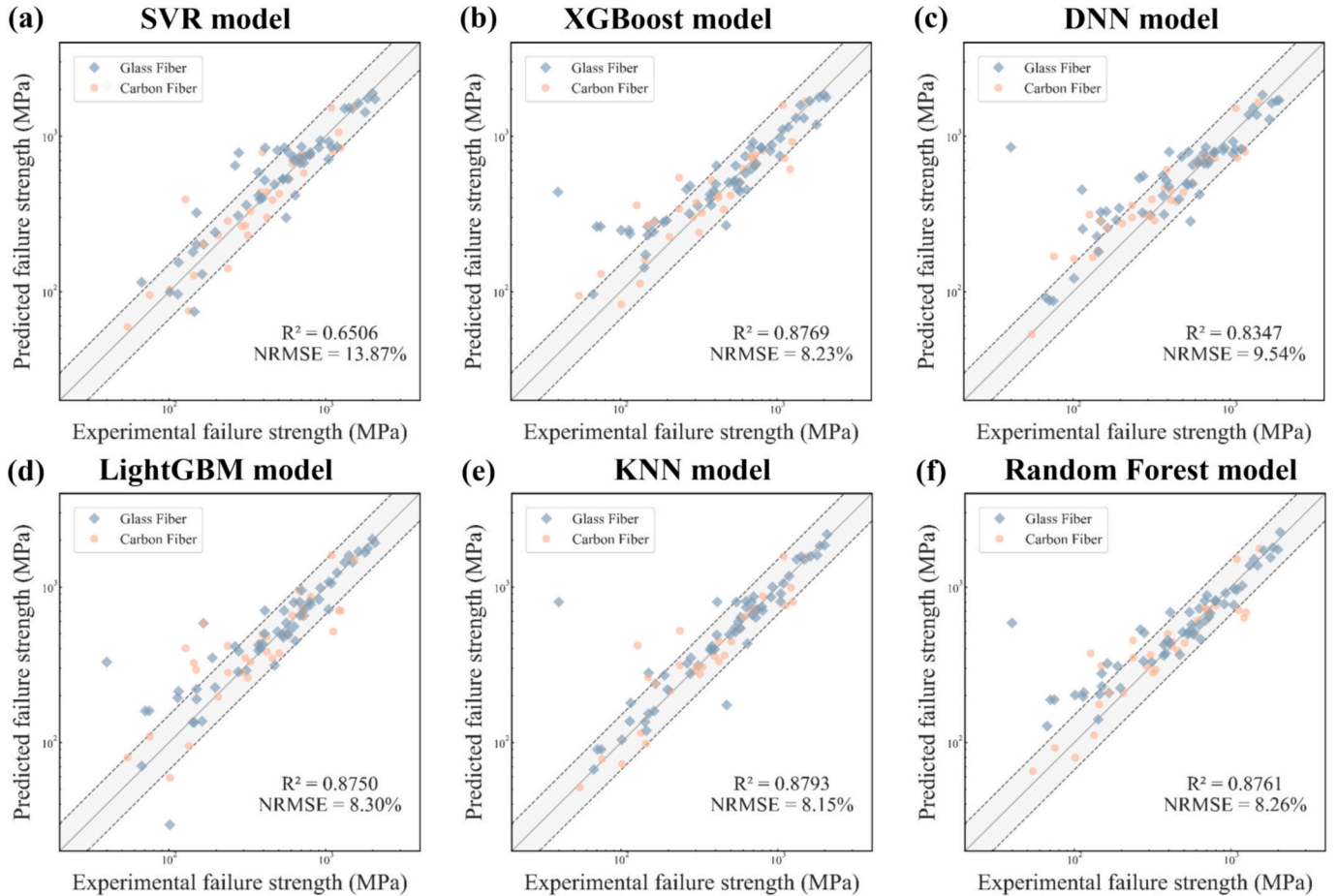


Fig. 12. Comparative benchmarking of predictive fidelity against representative machine-learning baselines. Scatter plots comparing the predicted versus experimental failure strengths for six representative data-driven baselines (SVR, XGBoost, DNN, LightGBM, KNN, and Random Forest).

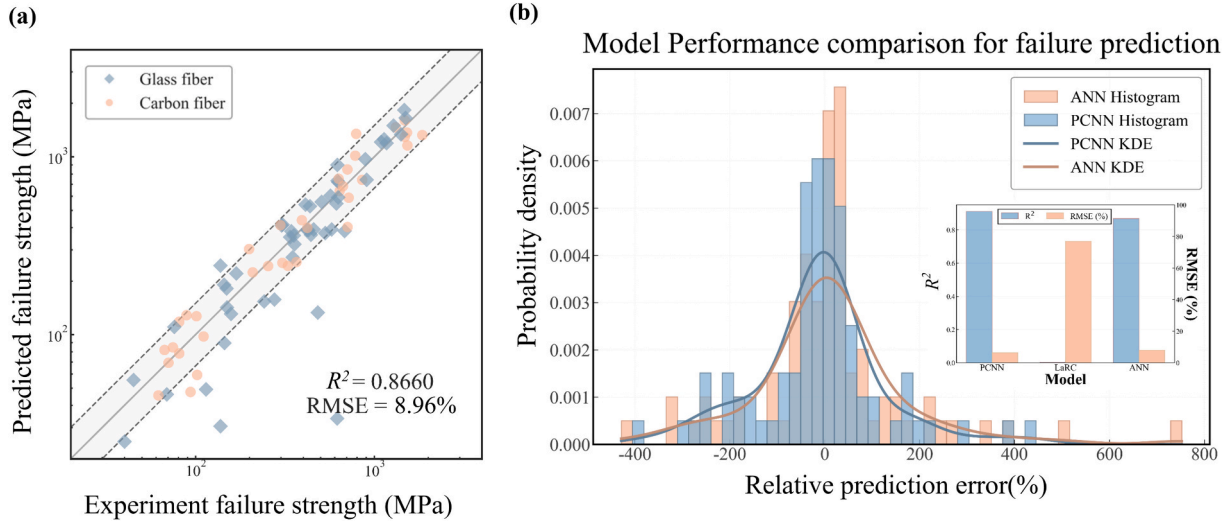


Fig. 13. Prediction performance of the baseline ANN and comparative error analysis. (a) Parity plot comparing experimental versus predicted failure strength on the hold-out test set, revealing outliers and higher dispersion in the absence of physical constraints. (b) The probability density analysis of relative prediction errors (right) visually confirms that the PCNN framework suppresses the heavy-tailed error distribution observed in the ANN model.

of the predicted values fall outside the factor of 1.5 error band. This phenomenon highlights the limitation of purely data-driven models when they lack guidance from physical prior knowledge: while the model statistically fits the main trend, its behavior at physical boundaries remains unstable. It may generate physically unreasonable interpolations to minimize global data error, especially in regions with sparse data or complex patterns.

The advantages of the PCNN framework are further confirmed by the statistical error analysis presented in Fig. 13b. The probability density curves reveal that the ANN model (orange line) exhibits a heavy-tailed error distribution, indicating its tendency to produce extreme prediction errors. By contrast, the PCNN (blue line) shows a narrower distribution highly concentrated near zero, demonstrating enhanced prediction consistency. This contrast underscores the key role of physical constraints: they improve precision while suppressing large errors by penalizing unphysical solutions, thereby enhancing model robustness.

The inset quantitative comparison in Fig. 13b further confirms the PCNN framework's better performance over the purely data-driven ANN model across all indicators: R^2 increased to 0.9198 and the NRMSE decreased to 6.36%. In contrast, the LaRC criterion serves as the source of physical constraints but shows low accuracy with an R^2 of approximately 0.05 when used as a predictor itself, exposes the limitations of rigid physical models in fitting complex real data.

To evaluate the model's stability under data-sparse conditions—a challenge in composite material characterization—we extended the ablation study by analyzing the performance evolution across varying training set ratios (from 10% to 50%). The results, summarized in the box plots of Fig. 14, reveal a divergent performance trajectory between the two models. As illustrated in Fig. 14a, the unconstrained ANN exhibits a performance collapse at the 10% training ratio. The median R^2 drops below 0.20, effectively indicating a failure to capture any meaningful constitutive trends. More critically, the vertical span of the box and whiskers for the ANN is exceptionally wide, reflecting stochastic volatility; this implies that the pure data-driven model is highly sensitive to the specific random subset of data used for training, leading to unpredictable outcomes. Conversely, the PCNN maintains a performance with a median $R^2 \approx 0.35$ and, a much tighter range. This compactness demonstrates that the physical constraints anchor the optimization, preventing the model from oscillating wildly due to data sparsity.

This stability advantage persists as the data volume increases. As shown in the RMSE analysis of Fig. 14b, while both models improve with more data, the ANN continues to exhibit larger error bars (whiskers) even at the 40%–50% ratios, indicating persistent local overfitting risks. In contrast, the PCNN achieves lower median errors and narrower variance. This analysis confirms that in regimes where the data gradient is noisy or weak, the physical gradient derived from the failure criteria provides a necessary regularization, ensuring that the model learns the

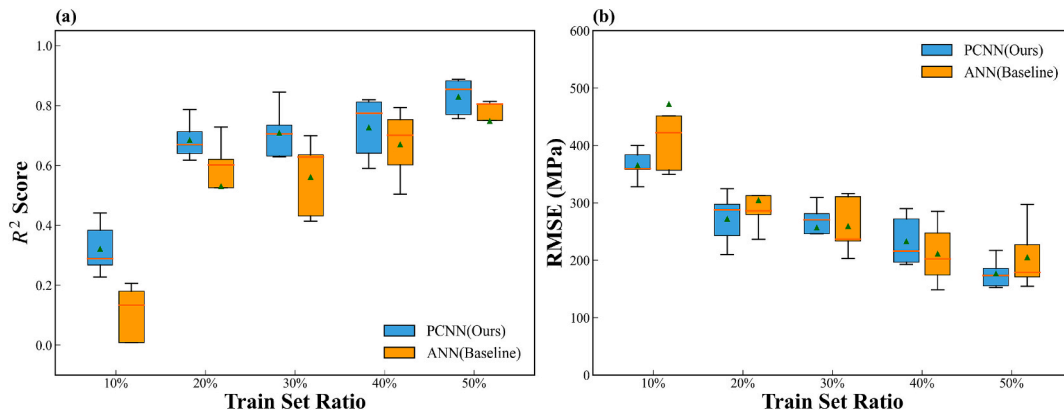


Fig. 14. Comparative assessment of model stability in data-sparse regimes. The statistical distribution of predictive performance is visualized via box plots for (a) the Coefficient of Determination (R^2) and (b) the Root Mean Squared Error (RMSE) across varying training set ratios (10%–50%).

underlying mechanics rather than merely memorizing sparse data points.

To further evaluate the model's generalization capability beyond data sparsity—specifically its ability to predict failure in entirely new material systems—we extended the ablation study using a Leave-One-Case-Out (LOCO) cross-validation strategy. In this experimental setup, specific test cases (e.g., Case 4, 6, and 7) were entirely excluded from the training set, forcing the model to predict the failure behavior of a “blind” material system based solely on knowledge learned from other cases. As illustrated in Fig. 15, the unconstrained ANN exhibits a marked performance degradation when extrapolating to these unseen domains. Specifically for Case 4, the ANN's Coefficient of Determination R^2 drops to 0.539, with predictions diverging from the experimental diagonal. This indicates that without physical guidance, the pure data-driven model tends to overfit the specific statistical distributions of the training set. In contrast, the PCNN maintains robust predictive fidelity, achieving an R^2 of 0.885 for Case 4 and outperforming the ANN in the highly scattered Case 7 ($R^2 = 0.582$ vs. 0.349). These results confirm that the physical constraints function as a critical inductive bias, anchoring the model to failure mechanisms and thereby ensuring consistent transferability even when direct experimental data is unavailable.

4.2.3. Effectiveness assessment of classical failure criteria

To evaluate the necessity of the PCNN framework, we first directly applied the classical failure criteria to the regression task. The LaRC, Tsai-Wu, and Cuntze criteria are applied directly to the test set to predict failure strength values, with the results visualized in the top row of Fig. 16 (panels a-c).

The evaluation results reveal dispersion between the predicted values of classical criteria and the experimental data. When used as direct regression models, these criteria produce numerous points that

deviate markedly from the ideal diagonal in Fig. 16. However, this discrepancy does not indicate flaws in the classical criteria themselves, but arises from a difference in objectives. These criteria are designed to define a physically self-consistent and continuous failure envelope in multidimensional stress space—a cornerstone of composite mechanics extensively validated for judging failure states. In contrast, the present task is the more demanding point regression problem of accurately predicting a specific failure strength scalar under a given loading path. While the deterministic forms of classical criteria provide a complete framework for smooth failure envelopes, their limited flexibility hinders precise point predictions. This contrast underscores the mechanism of the PCNN framework: rather than attempting to replace classical theories, PCNN instead extends them by transforming their embedded physical knowledge from a closed-form prediction rule into a flexible prior within a data-driven model.

4.3. Model mechanism and hyperparameter analysis

4.3.1. Loss convergence analysis

The impact of embedding different physical prior knowledge into the neural network is reflected in the training dynamics. This process is illustrated in the bottom row of Fig. 16 (panels d-f), which tracks the evolution of the total loss function for the three PCNN variants.

All models are driven solely by data loss during the initial warm-up phase (the first 800 epochs). The total loss decreases rapidly and tends to stabilize during this period. This indicates that the neural network has learned the primary non-linear mapping relationships and constructed a preliminary prediction model. A divergence in convergence behavior becomes apparent upon the introduction of the physics loss term. For the PCNN-TW and PCNN-Cuntze models, the total loss exhibits an upward trend, implying that the optimal solution derived from the initial data training encounters a certain degree of tension when attempting to align

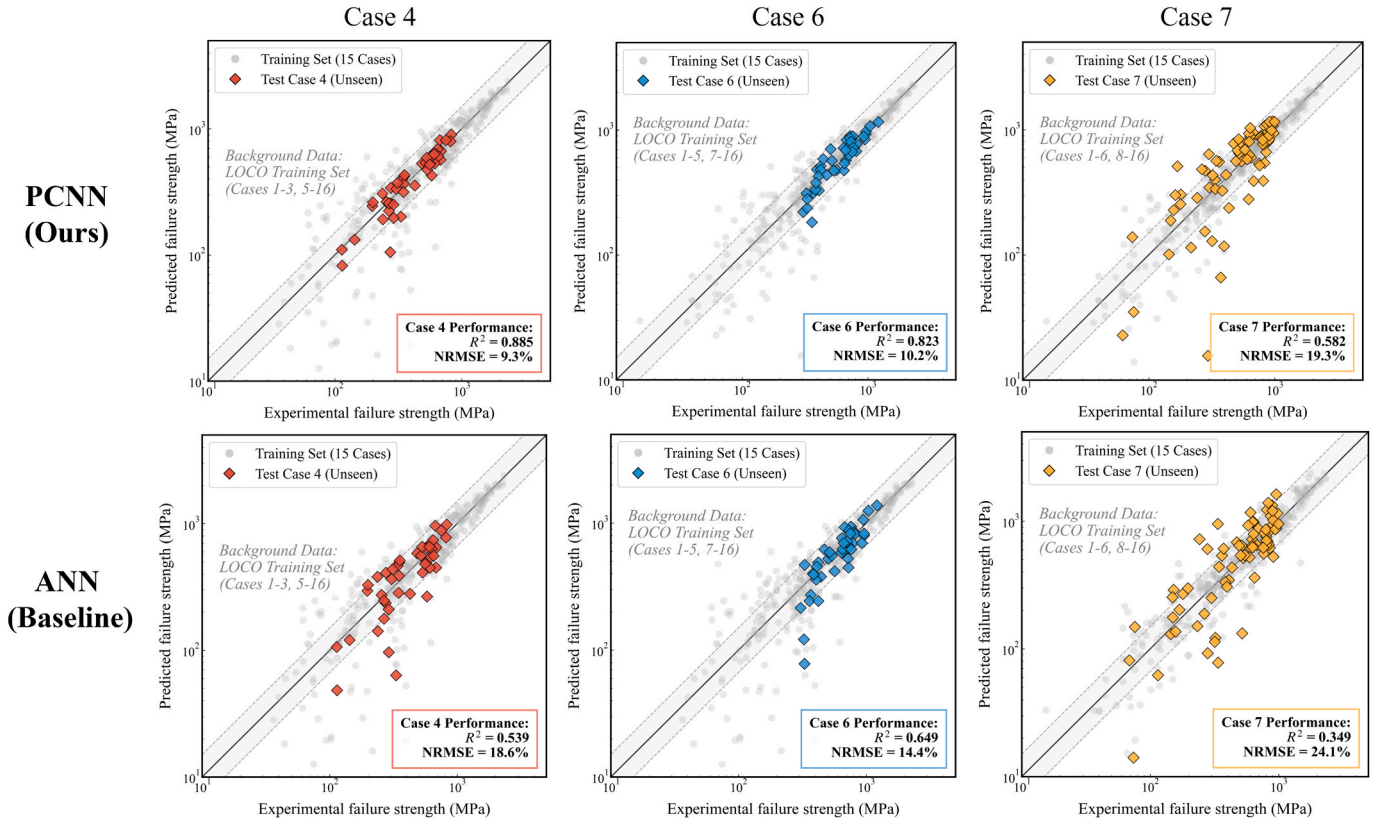


Fig. 15. Evaluation of generalization capability via Leave-One-Case-Out (LOCO) cross-validation. The parity plots contrast the predictive performance of the proposed PCNN (top row) against the baseline ANN (bottom row) on three representative unseen test cases (Case 4, 6, and 7). In each subplot, the grey background points represent the training data derived from the remaining cases, while the colored diamonds denote the specific unseen test data excluded from training.

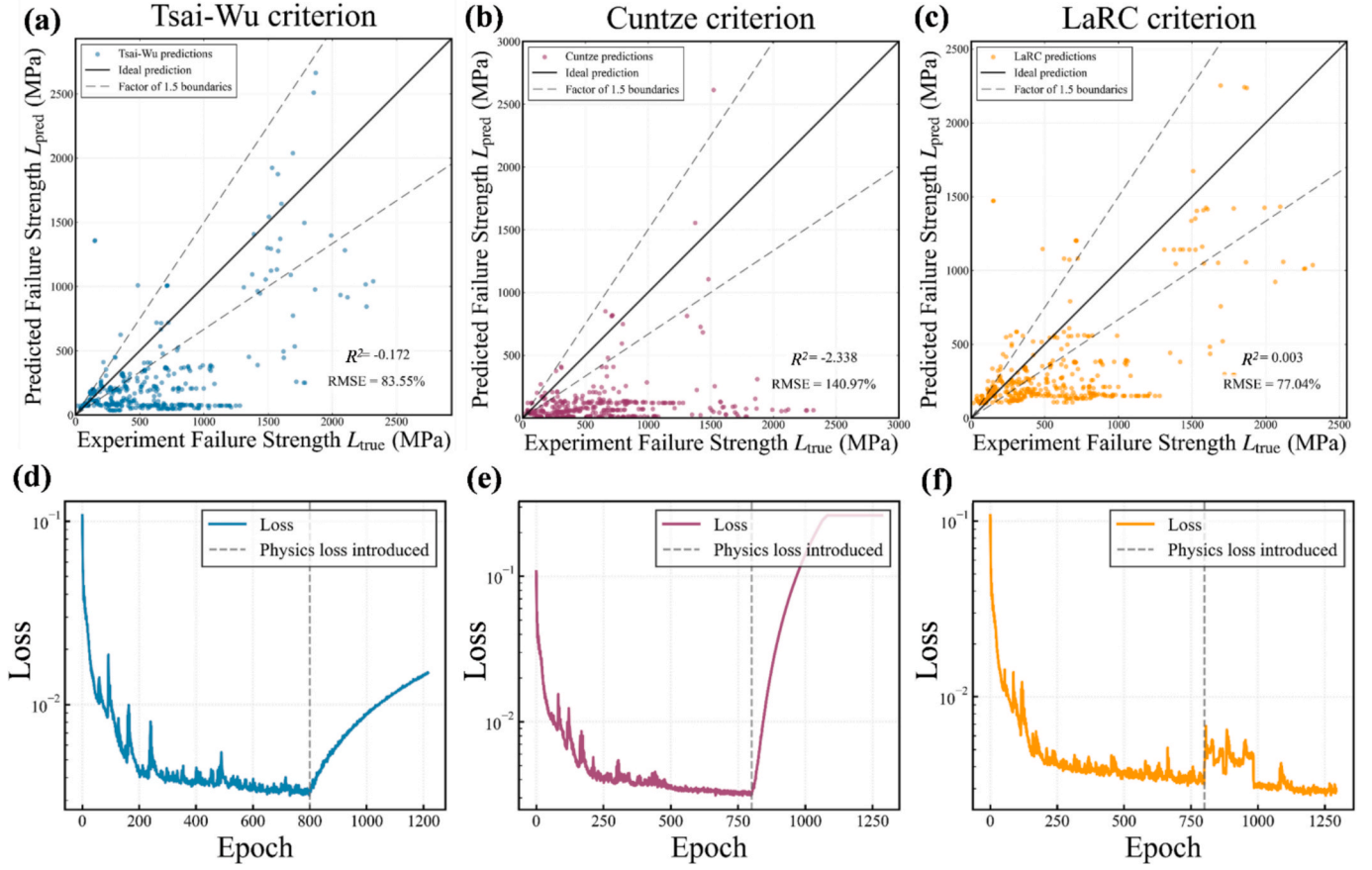


Fig. 16. Direct predictive capability of classical criteria and comparative training convergence. The top row (a-c) evaluates Tsai-Wu, Cuntze, and LaRC theories as standalone regression predictors, revealing their inability to capture experimental variance. The bottom row (d-f) illustrates the evolution of the total loss function for the corresponding PCNN variants. The vertical dashed line marks the introduction of physical constraints at epoch 800.

with the constraint boundaries defined by these criteria. Consequently, the training concludes via the early stopping mechanism. In comparison, the optimization trajectory of PCNN-LaRC demonstrates a more synergistic response; after a brief fluctuation, the total loss resumes a downward trend and eventually converges to a lower stable value. This behavior suggests that the physical prior knowledge provided by the

LaRC criterion is effectively compatible with the underlying failure patterns inherent in the experimental data, allowing the physical and data-driven components to optimize cooperatively.

4.3.2. Impact of physical weighting strategy on model performance

The weighting strategy for the physical loss term is a key factor

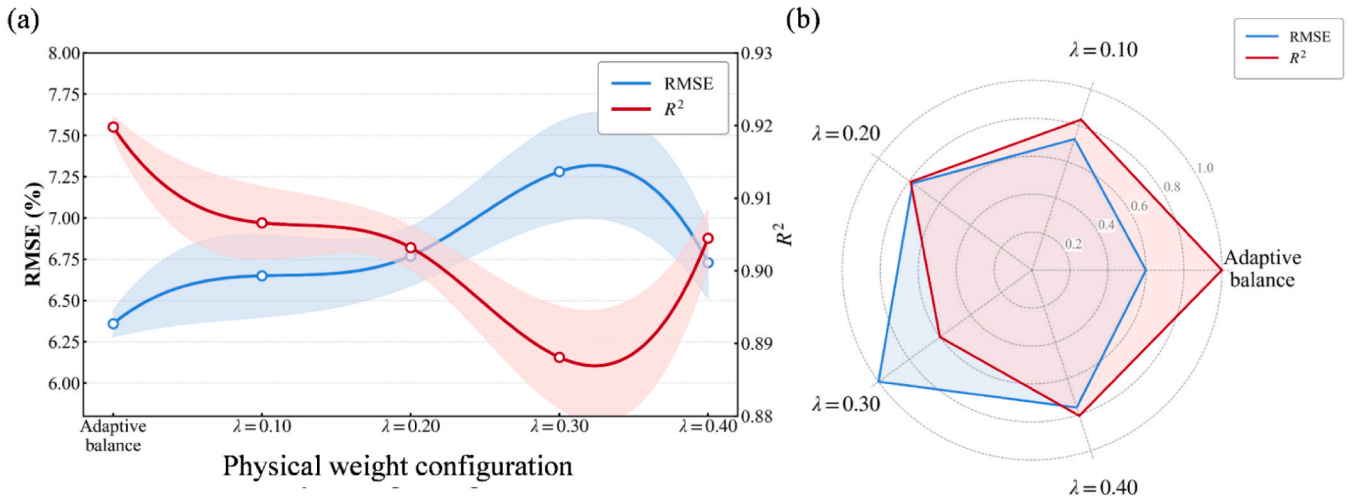


Fig. 17. Impact of physical loss weighting strategies on model prediction accuracy. (a) Dual-axis quantitative evaluation comparing the Normalized RMSE (blue) and Coefficient of Determination (R^2 , red line) achieved by the proposed Adaptive Balance mechanism versus static physical weights (λ). (b) Radar chart visualization highlighting the performance metrics across configurations.

determining the PCNN framework performance. This section evaluates the adaptive weighting mechanism against the traditional fixed weighting method to determine the optimal strategy.

The dual-axis plot in Fig. 17a provides a quantitative tracking of model performance across different configurations. The model employing the proposed adaptive weighting mechanism which dynamically regulates the balancing coefficient λ based on the loss target ratio α , achieves optimal values across all evaluation indicators. It obtains the highest coefficient of determination ($R^2 = 0.9198$) and the lowest Normalized Root Mean Square Error (NRMSE = 6.36%). The radar chart in Fig. 17b visually corroborates this dominance. The expansive area covered by the adaptive mechanism demonstrates its capacity to simultaneously maximize competing metrics, thereby achieving an optimal equilibrium that static weighting strategies fail to reach. This best performance is realized under the guidance of an optimized loss target ratio $\alpha = 0.1$.

In contrast, models adopting a fixed physical weight λ exhibit discernible performance limitations, highlighting the sensitivity of the optimization process to this hyperparameter. Specifically, the curves in Fig. 17a reveal a trade-off dilemma: as λ increases from 0.10 to 0.30, the prediction error rises, culminating in a performance trough at $\lambda = 0.30$ (NRMSE = 7.28%). This deterioration suggests that the high fixed weight imposes excessive rigidity, forcing the network to prioritize theoretical compliance over experimental data fidelity. Conversely, a static low weight may fail to provide sufficient regularization. The adaptive strategy circumvents this conflict by maintaining a dynamic gradient equilibrium between data corrections and physical constraints. This mechanism ensures that both components act synergistically throughout the training phase, thereby preventing either objective from disproportionately dominating the optimization process.

4.4. High fidelity of PCNN failure envelopes

Beyond its high accuracy in point prediction—quantitatively validated in previous sections—the PCNN framework offers a deeper advantage: its capability to generate high-fidelity and physically self-consistent failure envelopes. This capacity ensures that model outputs adhere to principles, offering value for engineering applications.

Accurately defining the failure envelope in a multidimensional stress space is fundamental to reliability design and safety assessment of composite structures. A desirable prediction model should not only reproduce the engineering failure frontier under different stress combinations, but also quantify the confidence boundaries associated with experimental dispersion. Fig. 18 presents the reconstructed failure envelopes for six representative cases (Case 1, 2, 4, 5, 7, and 8). The solid blue line and green band denote the final PCNN-LaRC Conservative Design Envelope and its 95% Confidence Interval, respectively, while the dashed curves denote representative analytical reference criteria (Puck, Cuntze, and Tsai-Wu) plotted for external visual benchmarking. To avoid ambiguity in the comparison, these analytical envelopes are evaluated using the same case-specific uniaxial strength values as those provided to the PCNN as material-property inputs.

The analysis of these results reveals that the model's predictive behavior exhibits distinct characteristics contingent upon the material system, addressing the challenge of universality in composite modeling. In carbon fiber systems characterized by high manufacturing consistency—specifically Case 2 (T300/BSL914C under $\tau_{xy} - \sigma_x$) and Case 8 (T300/Epoxy under $\tau_{xy} - \sigma_x$)—the PCNN demonstrates good agreement. As shown in Fig. 18(b) and 18(f), the generated failure envelopes are tightly confined with narrow confidence intervals (shaded regions), reflecting the low dispersion of the experimental data. Here, the Conservative Design Envelope (blue solid line) and the mean prediction virtually overlap, effectively reconstructing a smooth failure boundary

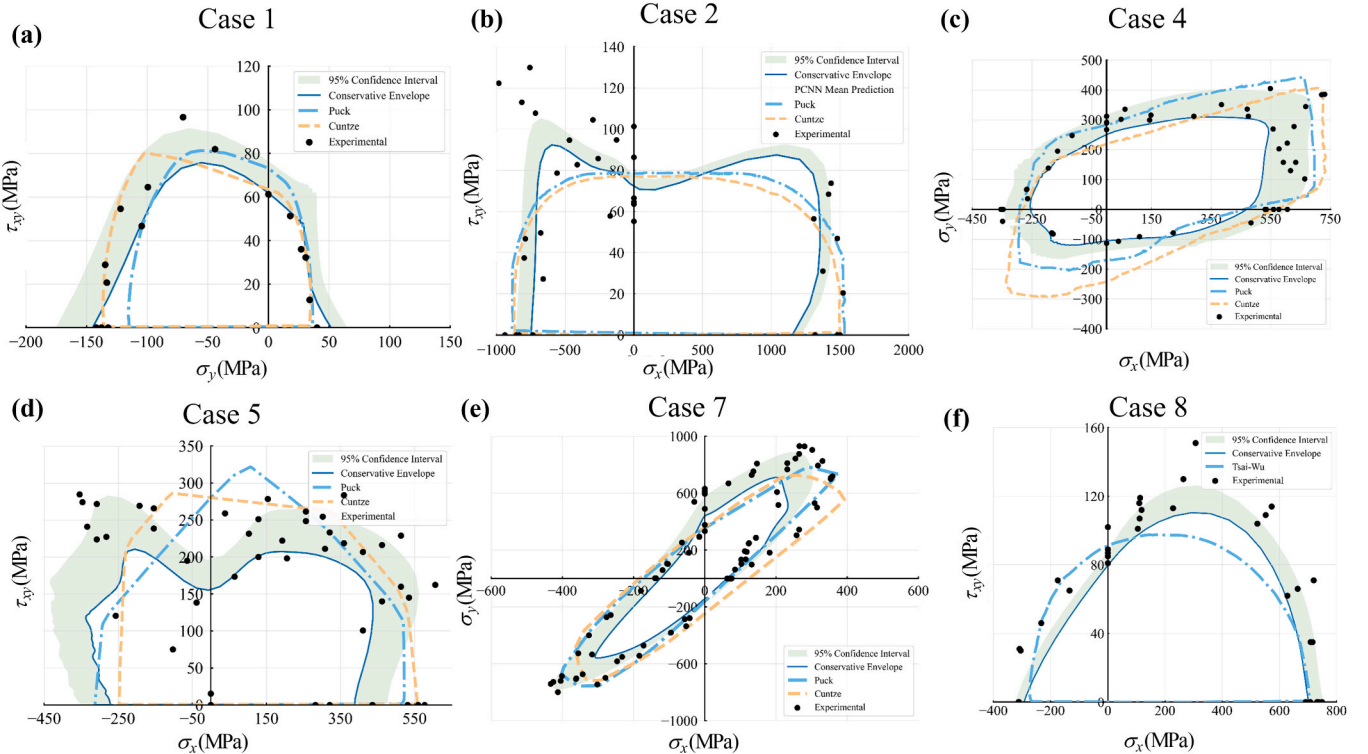


Fig. 18. Reconstructed failure envelopes with uncertainty quantification across diverse multiaxial stress states. Comparison of the final PCNN-LaRC Conservative Design Envelope (solid blue lines) and 95% Confidence Intervals (green shaded regions) against experimental data (black dots) and representative analytical reference criteria (Puck, Cuntze, and Tsai-Wu; dashed lines) for six representative cases. For each case, the dashed analytical envelopes are evaluated using the same case-specific uniaxial strength values as those used in the 9-dimensional material-property inputs to the PCNN, thereby ensuring material-property-level comparability.

that closely aligns with the clean experimental frontier.

Conversely, for E-glass/epoxy systems (Case 1, 4, 5, and 7), the experimental data exhibits pronounced scatter due to the inherent stochasticity of microstructural defects, including fiber waviness and variable interfacial bonding. In these high-variance regimes, a deterministic “high-fidelity” line is often physically ill-defined. The PCNN framework addresses this by capturing the associated predictive uncertainty through the confidence band. As observed in Case 1 ($\tau_{xy} - \sigma_y$, Fig. 18a) and Case 5 ($\tau_{xy} - \sigma_x$, Fig. 18d), rather than overfitting to local noise which would result in unphysical oscillations, the model projects a wider confidence interval. To translate this probabilistic output into actionable engineering guidance, we extract a Conservative Design Envelope, defined as the smoothed lower bound of the 95% confidence interval (bold blue line). Under the adopted envelope-level formulation, this lower-tail boundary corresponds to a smaller predicted failure magnitude along the same loading path and therefore represents a safety-oriented interpretation of the admissible domain. It is introduced here as a probabilistic safety margin for design-oriented assessment in the presence of experimental scatter, rather than as a universal deterministic reduction factor. Accordingly, in regimes with stronger dispersion or compression-dominated uncertainty, the lower bound may become intentionally more conservative, and should therefore be interpreted together with the mean prediction and the full confidence band.

Furthermore, the framework helps accommodate local discrepancies between classical criteria and experimental envelopes that can arise under complex multiaxial conditions. As evidenced in Fig. 18(c) for Case 4, the classical Puck and Cuntze criteria (dashed lines) exhibit deviations from the experimental data in the compression-compression quadrant, likely due to their underestimation of the strengthening effects from multi-axial constraints. If the PCNN were merely “smoothing” these theories, its prediction would collapse towards these biased theoretical boundaries. However, the PCNN envelope (solid blue line) diverges from the theoretical priors to accurately capture the experimental failure limits. This confirms that the framework successfully functions as a correction mechanism, leveraging the data-driven gradient to compensate for the epistemic uncertainty in the analytical priors.

To further clarify this correction mechanism in a representative and visually transparent manner, an additional diagnostic comparison for Case 4 is provided in Fig. 19, where the unconstrained ANN, the proposed PCNN, and a retained representative analytical prior (Cuntze) are

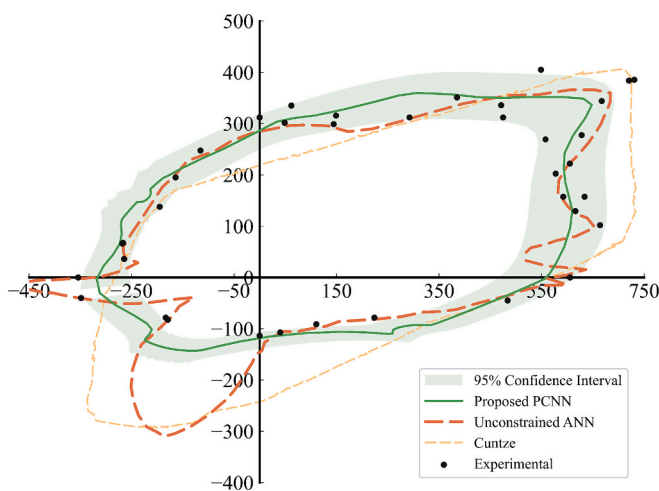


Fig. 19. Representative diagnostic comparison for Case 4 illustrating topology guidance and prior-bias correction. The unconstrained ANN, the proposed PCNN, and a retained representative analytical prior (Cuntze) are compared against the experimental envelope. Case 4 is highlighted because the compression–compression quadrant provides a stringent test of prior anchoring risk, where the retained analytical prior visibly deviates from the experimental boundary.

projected on the same failure-envelope plane. This case was selected deliberately because the compression–compression quadrant constitutes a stringent test of prior anchoring risk: in this region, the retained analytical prior exhibits a visible deviation from the local experimental frontier, making it possible to examine whether the PCNN merely smooths an existing theory or preserves sufficient flexibility to depart from it when the data indicates otherwise. As shown in Fig. 19, the unconstrained ANN exhibits a less stable local geometric evolution, with stronger waviness and more pronounced turning fluctuations, whereas the PCNN maintains a more coherent and physically interpretable boundary while remaining closer to the experimental envelope in the compression-dominated region. At the same time, the PCNN does not collapse onto the retained analytical prior in the quadrant where that prior is locally biased, but instead preserves a clear data-driven deviation toward the experimentally supported boundary. Quantitatively, in the compression–compression quadrant, the local RMSE of the PCNN boundary relative to the experimental envelope is 43.6 MPa, compared with 147.2 MPa for the unconstrained ANN and 182.7 MPa for the retained analytical prior. This result indicates that the physical loss does not rigidly anchor the network to a biased analytical manifold; rather, in the present envelope-reconstruction setting, it acts as a topology-guidance mechanism that suppresses non-physical local artifacts while still allowing data-driven correction where experimental evidence is available. Here, the retained Cuntze curve is used only as a representative analytical prior for this topology-guidance test, while the probabilistic confidence band is preserved for consistency with the uncertainty-aware formulation discussed above. The observed inefficiency of traditional criteria in direct prediction stems not from theoretical flaws but from their inherent inability to capture the high-dimensional, nonlinear deviations introduced by real-world material variability. Within this framework, physical criteria anchor a consistent subspace, while the neural network learns the systematic residuals between this physical baseline and the experimental response. This approach—using physics as a guide and data for refinement—helps maintain physical consistency even in multi-axial regimes where data are sparse or scattered.

4.5. Interpretability and safety assurance

While Sections 4.1–4.4 focused on predictive fidelity, stability, and envelope reconstruction with uncertainty bounds, it is also useful to discuss how the proposed framework can be interpreted and assessed from an engineering perspective. In this section, we summarize the interpretability position of the PCNN as a physically constrained grey-box and present complementary evidence for traceability at two levels: output-level auditability through criterion-referenced checks, and input-level attribution consistent with the physically designed feature groups.

4.5.1. Physically-constrained grey-box rationale and auditability

To clarify the interpretability scope of the proposed framework, we acknowledge that the PCNN operates as a “physically-constrained grey-box” system [39], distinct from the fully transparent “white-box” nature of classical analytical criteria. In analytical models such as LaRC04 or Cuntze, parameters (e.g., transverse shear strength or a friction coefficient) possess direct physical definitions derived from mechanistic assumptions. In contrast, the internal weights and biases of the neural network component in our PCNN serve as abstract feature representations and do not directly correspond to specific material properties. This distinction is important for interpretation: while analytical criteria provide parameter-level transparency and well-established physical meaning, their practical use at the engineering envelope level may still involve simplifying assumptions and can exhibit discrepancies under complex multiaxial stress states.

The retention of a neural component is therefore motivated by the need to accommodate nonlinear residual relationships that are difficult

to capture fully in closed-form expressions. Composite failure may involve coupled effects and regime transitions that are challenging to formalize without introducing additional idealizations. By leveraging the approximation capability of deep neural networks [62], the PCNN can learn data-supported deviations from the idealized analytical envelope while preserving the physically meaningful structure provided by classical criteria. In this sense, the data-driven component is intended to complement—not replace—classical theories, improving envelope reconstruction when the available experimental evidence indicates departures from idealized assumptions.

However, to mitigate the opacity inherent in data-driven components for safety-critical applications, this study adopts the “Physics-Informed” paradigm [41] to shift the focus of verification from internal parameter interpretability to output functional consistency. In the proposed PCNN, the “physics” is applied at the loss-function level, and this is precisely why the framework is not claimed to be white-box. A practical implication is that the prediction can be audited through a physically meaningful and reproducible chain: the network outputs a scalar magnitude L , which is combined with the direction vector $\mathbf{d}$ to reconstruct the full stress tensor; the reconstructed stress is then evaluated by established failure criteria (e.g., LaRC04, Cuntze) to compute failure indices and identify the controlling mode(s). These criterion-based quantities provide externally interpretable diagnostics that can be reported alongside the predicted envelope, enabling reviewers and practitioners to verify whether the predicted boundary remains consistent with the intended failure-mode structure and to detect physically implausible artifacts that may arise from unconstrained regression. Importantly, because the criteria enter as soft penalties during training, this procedure should be understood as providing an additional, physics-referenced consistency check and an auditable basis for model assessment, rather than a formal proof of laminate-scale mechanistic completeness or a certification guarantee. This perspective is broadly consistent with “learning assurance” discussions in the aerospace community [63], in that it emphasizes verifiable model behavior at the output level while acknowledging the intrinsic limitations of grey-box neural components. While this criterion-referenced procedure supports output-level auditing, it does not directly indicate how the trained network allocates reliance among material, laminate, and loading descriptors. We therefore complement it with a grouped SHAP analysis for input-level traceability in Section 4.5.2.

4.5.2. Mechanistic attribution via grouped SHAP

We next examine the input-level traceability of the trained PCNN through a grouped attribution analysis. As introduced in Section 2.2.1, the 20-dimensional input vector is organized into three physically meaningful information sources—material properties (9D), laminated architecture (5D), and stress ratio (6D)—which naturally supports interpretation at the level of mechanism-relevant feature groups. The purpose of this analysis is not to claim causality, but to provide a transparent summary of how the trained model allocates predictive reliance among material thresholds, laminate descriptors, and loading-path descriptors under different loading regimes, thereby strengthening the overall safety-assurance narrative in a manner consistent with the decoupled prediction setting $\sigma = L\mathbf{d}$.

We employ Shapley additive explanations (SHAP) to quantify the attribution of each input feature to the scalar prediction $L = f(\mathbf{x})$. In the SHAP framework, the prediction is decomposed as

$$f(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})] + \sum_{i=1}^M \phi_i(\mathbf{x}), \quad (38)$$

where M is the number of input features and $\phi_i(\mathbf{x})$ denotes the Shapley value of feature i . The Shapley value is defined by the standard coalition-based formulation

$$\phi_i(\mathbf{x}) = \sum_{S \subseteq \mathcal{F} \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_{S \cup \{i\}}(\mathbf{x}) - f_S(\mathbf{x})] \quad (39)$$

with $\mathcal{F}$ the full feature set and f_S the model evaluated with the subset S . To align with the physically designed feature space, we aggregate SHAP magnitudes within each group

$$A_g = \frac{1}{N} \sum_{n=1}^N \sum_{i \in g} |\phi_i(\mathbf{x}^{(n)})|, \quad P_g = \frac{A_g}{\sum_g A_g} \times 100\% \quad (40)$$

where N is the number of samples in a given loading-mode subset. In addition, for the stress-ratio components (6D), we report the feature-wise mean absolute attribution $\frac{1}{N} \sum_{n=1}^N |\phi_i(\mathbf{x}^{(n)})|$ and visualize the corresponding SHAP distributions colored by the normalized feature value, enabling a compact assessment of regime-dependent attribution patterns.

Fig. 20 indicates that the relative contributions of the three feature groups vary systematically with the loading regime. Under SX-compression-dominant conditions, the 9D material property group accounts for the largest share of the total attribution (approximately one-half of the grouped SHAP budget), whereas the 5D laminate architecture and 6D stress-ratio groups contribute smaller but non-negligible shares. In contrast, under SY-tension-dominant conditions, the grouped contribution shifts toward a more balanced partition in which the 5D laminate architecture share becomes comparable to (or slightly exceeds) the 9D material share, suggesting that structural descriptors (e.g., stacking sequence and symmetry) play a more prominent role when the response is governed by transverse or off-axis effects. For shear-dominant conditions, the attribution is distributed across all three groups with a modest increase in the stress-ratio share, consistent with the expectation that shear-driven failure envelopes are sensitive not only to shear-related allowable (material thresholds) but also to laminate-level anisotropy and loading-path composition. Importantly, these observations should be interpreted as model attributions within the learned mapping, rather than direct causal statements; nevertheless, the regime-dependent redistribution across (material, laminate, loading) groups aligns with the mechanistic intent of the feature construction.

The stress-ratio attribution patterns (Rows 2–3) further refine the regime-wise interpretation by identifying which stress-direction components most strongly influence the predicted failure magnitude L . In SX-tension-dominant cases, the attribution is primarily associated with the axial component, whereas in shear-dominant cases the in-plane shear component exhibits the largest mean $|\text{SHAP}|$, and its SHAP distribution is largely one-signed across the plotted samples. This one-signed tendency suggests a consistent directional effect of increasing shear participation on the predicted failure scaling, which is compatible with a stable shear-triggered response in the learned mapping. In compressive regimes, the comparatively elevated attribution associated with the out-of-plane normal component (relative to the pure axial component) indicates that the PCNN is not solely relying on a single in-plane descriptor; instead, it appears to account for additional three-dimensional coupling embedded in the training data and the physics-constrained learning objective. Such behavior is consistent with the motivation of retaining a neural residual component to represent coupled effects and regime transitions that are difficult to capture by closed-form criteria alone, while still enforcing physically admissible outputs through the loss-level constraints and stress reconstruction pathway.

Overall, the grouped SHAP analysis provides an interpretable summary of how the PCNN leverages the three physically designed input information sources across loading regimes, complementing (rather than replacing) the criterion-based output auditing chain emphasized in Section 4.5. The analysis supports the view that the proposed framework operates as a physically constrained grey-box: the internal weights remain abstract, yet the decision logic can be examined at the level of

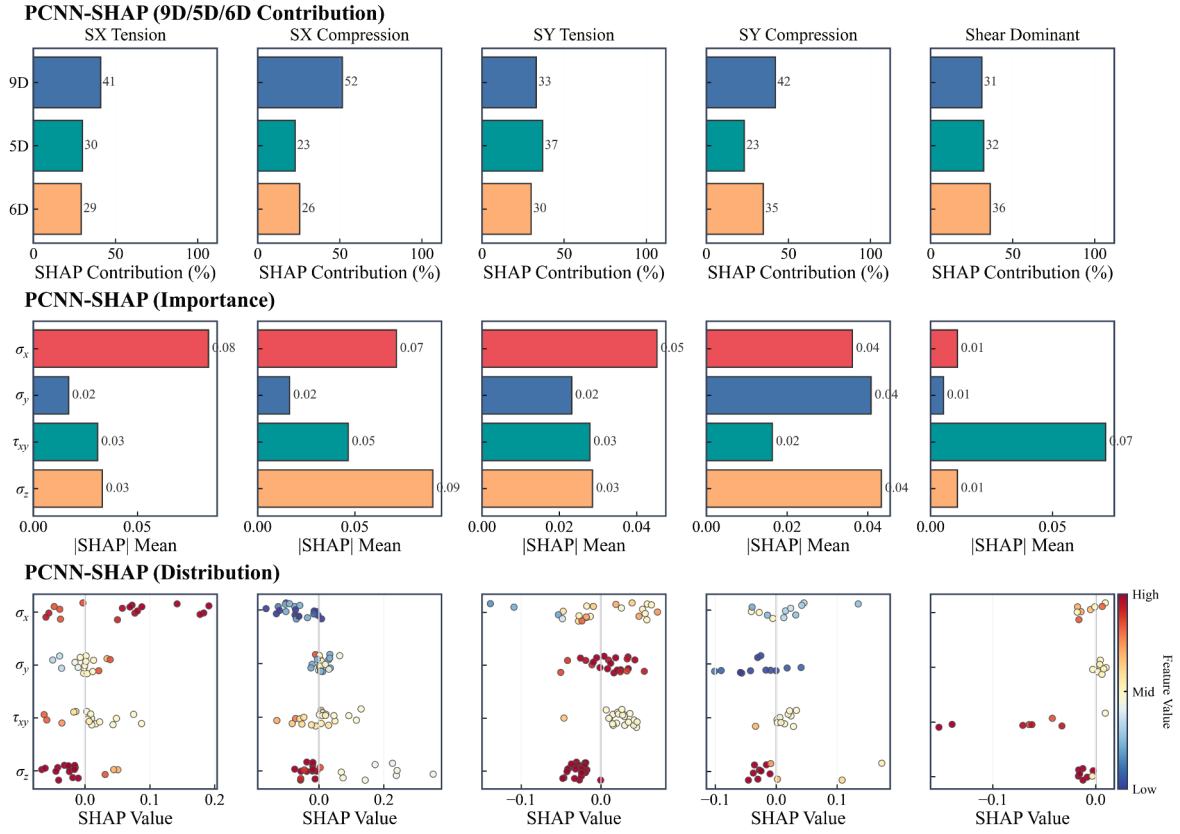


Fig. 20. Mechanistic attribution analysis of the PCNN using grouped SHAP values across loading-mode subsets. The top row reports the normalized contribution shares of the three physically defined feature groups (material properties 9D, laminated architecture 5D, and stress ratio 6D). The middle row shows the mean absolute SHAP values for representative stress-ratio components. The bottom row illustrates the corresponding SHAP value distributions, where point colors indicate the normalized feature magnitude.

physically meaningful feature groups, and the outputs can be audited through stress reconstruction and established failure indices. Two practical limitations should be noted when interpreting these plots: SHAP reflects attribution within the trained model and does not by itself establish causality, and correlations among material/laminate descriptors may redistribute attribution across features within a group. For these reasons, we emphasize group-level trends and regime-consistent patterns, and treat the SHAP evidence as a complementary diagnostic that strengthens the overall traceability and safety-assurance narrative representative machine-learning baselines.

5. Conclusions and remarks

This study presents a Physics-Constrained Neural Network (PCNN) framework to address the challenge of predicting composite material failure under complex, multi-axial stress states. By effectively integrating classical failure criteria with deep learning, the framework reconciles the flexibility of data-driven models with the rigor of physical laws, thereby overcoming the interpretability limitations of purely black-box approaches while surpassing the predictive rigidity of analytical criteria alone. For clarity, Tsai-Wu, Cuntze, and LaRC in the following summary refer to the three candidate physical priors embedded in the PCNN variants, whereas the representative envelope plots in Fig. 18 additionally include Puck, Cuntze, and Tsai-Wu as external analytical references for visual benchmarking. The main contributions of this study are summarized as follows:

1. Developed framework with adaptive balancing: This study establishes a PCNN framework that integrates a decoupled prediction

strategy with an adaptive weighting mechanism. This architecture effectively resolves the gradient imbalance between physical and data losses, ensuring that physical constraints actively guide the optimization process without hindering the model's convergence on complex experimental data.

2. Quantifying the impact of physical fidelity: We demonstrate a definitive positive correlation between the fidelity of inductive bias and predictive accuracy. The micromechanics-based PCNN-LaRC model outperformed macroscopic (Tsai-Wu) and simplified (Cuntze) variants, achieving the highest accuracy ($R^2 = 0.9198$). This suggests that specialized domain knowledge serves as a regularizer compared to generic constraints, particularly in data-sparse regimes.
3. High-fidelity reconstruction of failure envelopes: The framework reconstructs physically consistent failure envelopes, thereby correcting the local biases inherent in traditional analytical criteria. By effectively suppressing the heavy-tailed error distribution of pure data-driven models, the PCNN delivers reliable, closed boundaries for multi-axial stress states, offering a practical tool for safety-critical composite design.

Beyond its quantitative accuracy, the PCNN framework demonstrates that embedding high-fidelity physical constraints (e.g., LaRC) rectifies data-driven model behavior. By establishing a synergistic relationship—where physics anchors a consistent baseline and the network learns real-world deviations—the method corrects local biases in classical criteria and suppresses the heavy-tailed errors of purely data-driven models. The resulting physically consistent, data-calibrated failure envelopes provide a more reliable tool for safety-critical design in fields like aerospace, classical criteria serve as respected references and

structured constraints, while the network accommodates residual effects indicated by data.

Notwithstanding the observed stability of the probabilistic framework, a fair assessment reveals inherent methodological limitations that merit objective discussion. First, the framework's capacity to calibrate classical theories is data-contingent. While the adaptive weighting mechanism allows the model to rectify theoretical deviations (as observed in the compression-compression quadrant of Case 4), this correction capability relies on the availability of local experimental evidence to drive the data-driven gradient. In stress states where experimental data is extremely sparse or absent, the optimization landscape is naturally guided by the physical loss to ensure convergence. Consequently, in these unexplored regimes, the PCNN prudently aligns with the theoretical trends of the physical prior. This behavior ensures that the model maintains a conservative and physically consistent baseline, avoiding the risk of unphysical extrapolation often seen in purely data-driven approaches while respecting the established mechanics of the chosen criterion.

Second, the elevation of safety through Uncertainty Quantification (UQ) introduces a trade-off in both computational efficiency and design conservatism. The adoption of the Monte Carlo Dropout inference strategy requires multiple stochastic forward passes (e.g., $N = 100$) to generate reliable confidence intervals for a single stress state. Compared to the instantaneous computation of deterministic analytical criteria or standard neural networks, this probabilistic inference introduces a substantial computational overhead. This increased latency poses a potential bottleneck for large-scale engineering applications, particularly within iterative finite element simulations where the failure criterion must be queried millions of times, highlighting a need for more efficient uncertainty propagation methods in future iterations. In addition, the lower confidence bound used in the present study should be interpreted as a probabilistic safety margin for envelope-level assessment under the adopted proportional-loading setting, rather than as a universal deterministic knock-down factor. In regions characterized by larger uncertainty, this lower-tail estimate may become intentionally more conservative, which improves risk control but may reduce economic efficiency if used in isolation.

Future work will focus on extending the framework's capability from the current "Initial Failure" prediction to "Progressive Damage Analysis." Specifically, by integrating continuum damage mechanics (CDM)

based degradation laws into the physics-constrained loss function, the model aims to capture the non-linear post-peak softening behavior and stress redistribution inherent to composite laminates. This extension will allow for the characterization of the complete stress-strain response beyond the linear elastic limit. Parallel to this physical expansion, we intend to address the computational overhead associated with the current probabilistic inference. Future research will explore model distillation or variational inference techniques to construct a lightweight surrogate model. The objective is to retain the safety benefits of Uncertainty Quantification while reducing the inference latency to a level compatible with the efficiency requirements of iterative finite element simulations.

CRediT authorship contribution statement

Ruizheng Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dazhi Zhao:** Writing – review & editing, Visualization, Software, Methodology, Conceptualization. **Peng Zhang:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Keke Tang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.: Mathematical formulations of failure criteria

A.1. Tsai-Wu Criterion Parameters

The strength tensor coefficients in the Tsai-Wu criterion are determined from uniaxial strengths. The linear coefficients are defined as:

$$F_1 = \frac{1}{X_T} - \frac{1}{X_C}; F_2 = \frac{1}{Y_T} - \frac{1}{Y_C}; F_3 = \frac{1}{Z_T} - \frac{1}{Z_C} \quad (A.1)$$

The quadratic coefficients are given by:

$$F_{11} = \frac{1}{X_T X_C}; F_{22} = \frac{1}{Y_T Y_C}; F_{33} = \frac{1}{Z_T Z_C} \quad (A.2)$$

$$F_{44} = \frac{1}{S_{23}^2}; F_{55} = \frac{1}{S_{13}^2}; F_{66} = \frac{1}{S_{12}^2}$$

Interaction term coefficients:

To determine the interaction term coefficients, a normalized parameter is typically introduced.

$$F_{12} = -\frac{1}{2}\sqrt{F_{11}F_{22}}; F_{13} = -\frac{1}{2}\sqrt{F_{11}F_{33}}; F_{23} = -\frac{1}{2}\sqrt{F_{22}F_{33}} \quad (A.3)$$

Here, X , Y , and Z represent the strengths in the principal material directions 1, 2, and 3. The subscripts T and C denote tensile and compressive properties, respectively. S_{12} , S_{13} and S_{23} represent the shear strengths in the respective principal planes.

A.2. Cuntze Criterion: Fiber Failure Mode.

While the text presents the Inter-Fiber Failure (IFF) formulation, the Cuntze criterion evaluates Fiber Failure (FF) as an independent mode driven by longitudinal stress σ_1 . The failure effort is defined as:

$$\left(\frac{\sigma_1}{R_1}\right)^2 = 1 \quad (\text{A.4})$$

Here, R_1 is the tensile strength X_T , or the compressive strength X_C when σ_1 is negative.

The Inter-Fiber Failure mode assumes that failure occurs on a potential failure plane parallel to the fiber direction. First, the global stress tensor is transformed into a local coordinate system defined by a fracture angle θ . The local stress components on this plane are the normal stress (σ_n), longitudinal shear stress (τ_l), and transverse shear stress (τ_t):

$$\begin{cases} \sigma_n = \frac{\sigma_2 + \sigma_3}{2} + \frac{\sigma_2 - \sigma_3}{2} \cos(2\theta) + \tau_{23} \sin(2\theta) \\ \tau_l = \tau_{12} \cos(\theta) - \tau_{13} \sin(\theta) \\ \tau_t = -\frac{\sigma_2 - \sigma_3}{2} \sin(2\theta) + \tau_{23} \cos(2\theta) \end{cases} \quad (\text{A.5})$$

A.3. LaRC04 Criterion: Additional Failure Modes

The LaRC04 criterion encompasses four distinct failure modes. Beyond the matrix compression mode detailed in the main text, the remaining three modes are defined as follows.

Under tensile load ($\sigma_x \geq 0$), failure is determined by the maximum stress criterion, with the failure index I_F defined as:

$$I_F = \frac{\sigma_x}{X_T} = 1 \quad (\text{A.6})$$

Here, X_T represents the uniaxial tensile strength of the material in the fiber direction. Under compressive load ($\sigma_x < 0$), the criterion accounts for fiber kinking instability defined as:

$$I_F = \frac{-\sigma_x}{X_C - \frac{X_C^2 / 4G_{12}}{1 - X_C / 2G_{12}}} = 1 \quad (\text{A.7})$$

In this equation, X_C is the compressive strength in the fiber direction, and G_{12} is the in-plane shear modulus of the material.

The matrix failure mode is the core innovation of the LaRC criterion. When the normal stress is compressive ($\sigma_n < 0$), the criterion incorporates the transverse compression contribution:

$$I_M = \left(\frac{\tau_T}{S_T - \eta_T \sigma_n}\right)^2 + \left(\frac{\tau_P}{S_P - \eta_P \sigma_n}\right)^2 + \left(\frac{\sigma_n}{Y_C}\right)^2 = 1 \quad (\text{A.8})$$

In this case, S_T and S_P are the transverse and longitudinal (parallel to fibers) shear strengths, while η_T and η_L are friction coefficients, Y_C is the transverse compressive strength of the matrix.

Data availability

The code is publicly available at <https://github.com/ruiheng04/Physics-Constrained-Neural-Networks-PCNN> for non-commercial academic use.

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